

From Source to Execution: Design and Implementation of a Statically-Typed Programming Language with Type Inference

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ABSTRACT

This report presents the design and implementation of a small, statically-typed interpreted programming language realised through a four-stage compiler pipeline. The language provides four primitive types — Integer, Float, Boolean, and String — and supports arithmetic and equality expressions, variable assignment with type inference from first use, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` function. Types are inferred from program context rather than declared explicitly, combining static type safety with a declaration-free syntax. The pipeline consists of a hand-written lexer, a recursive-descent parser that constructs an abstract syntax tree, a static type checker that performs inference and rejects ill-typed programs before execution, and a tree-walking interpreter. Both a command-line runner and an interactive desktop interface expose each pipeline stage for inspection. The report documents the formal grammar, the type inference algorithm, the system architecture, and the automated test suite of 62 tests that validates the complete implementation.

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CHAPTER 1

INTRODUCTION

Programming languages define the formal contract between a programmer’s intent and machine execution. Building a language processor from source text through to evaluated output requires a coherent pipeline of components, each of which must correctly transform its input before passing it downstream. Lexical analysis identifies atomic tokens; parsing imposes hierarchical structure; static type checking validates semantic correctness before execution; and interpretation evaluates the validated program. When these stages are connected cleanly, errors are caught at the earliest possible point, and the overall system remains tractable to reason about, test, and extend.

This report presents the design and implementation of a small statically-typed interpreted language that realises this pipeline in full. The language supports four primitive types — Integer, Float, Boolean, and String — and provides arithmetic and equality expressions, assignment, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` statement. Types are inferred from program context rather than declared explicitly, combining the safety of static typing with the conciseness of a declaration-free syntax. The implementation spans three principal components developed in pipeline order: the lexer and symbol table, the recursive-descent parser and abstract syntax tree, and the type checker with interpreter. Each component was completed before the next was begun, since each stage consumes the structured output of its predecessor.

The implemented system can be invoked from the command line or through an interactive desktop interface, both of which surface the four pipeline stages — tokens, AST, inferred type table, and execution output — for inspection. An automated regression suite of 62 tests verifies correctness across all stages and language features.

The remainder of this report is organised as follows. Chapter 2 presents the language design, token specification, grammar, and operator precedence rules. Chapters 3 and 4 cover the front-end implementation in pipeline order: lexical analysis and the symbol table, then recursive-descent parsing and the AST. Chapter 5 explains the type inference algorithm and the semantic rules used by the checker. Chapter 6 presents the system architecture and implementation,

including the end-to-end pipeline, project layout, and technology stack. Chapter 7 discusses testing and verification, and Chapter 8 concludes the report.

CHAPTER 2

LANGUAGE DESIGN AND GRAMMAR

2.1 Types

The language provides four primitive types, chosen to cover the core domains of numeric computation, logical control flow, and text manipulation while keeping the type system tractable and the semantics well-defined.

2.1.1 *Integer*

The Integer type represents whole numbers such as 0, 7, and 42. Integer literals are written as one or more decimal digits without quotation marks or a decimal point.

2.1.2 *Float*

The Float type represents real-valued numbers written with a decimal point, such as 3.14 or 0.5. Float values are used exclusively with the dot-suffixed arithmetic operators (+., -., *., /.), following the Caml Light convention. Integer division (/) performs floor division and returns Integer: for example, 10 / 3 evaluates to 3 : Integer.

The language intentionally does not perform implicit numeric promotion between Integer and Float values. Mixed-type arithmetic such as 1 + 2.5 is rejected by the type checker. Instead, the language follows the Caml Light convention of using separate operator tokens for integer and floating-point arithmetic.

2.1.3 *Boolean*

The Boolean type represents logical truth values. The two Boolean literals are true and false. Boolean values are used in conditions and as results of comparison expressions.

2.1.4 *String*

The String type represents sequences of characters enclosed in double quotes, for example "hello". Strings can be assigned to variables and printed through the built-in print() function. The surrounding quotes are stripped by the parser when the Literal node is created, so

the value stored internally is the bare character sequence without surrounding quotes. At run-time, the built-in `print()` function re-adds double quotes around string values, so `print("plc")` outputs `"plc" : String`.

2.1.5 *Void*

Although the language exposes exactly four programmer-visible types, the implementation introduces a fifth value, `Void`, within the `LanguageType` enumeration. Its sole purpose is to provide a well-defined return type for functions that contain no `return` statement, allowing the symbol table to record a consistent entry for every function regardless of whether it produces a value. `Void` cannot appear in source code and is never visible to the programmer; it is a bookkeeping device used internally by the type checker. Therefore, `Void` does not extend the programmer-visible type system required by the assignment.

2.2 Static Typing

The language employs static typing, meaning that type errors are detected at type-checking time rather than at runtime. No explicit variable type declarations are required. Instead, the type checker infers types from literals, assignments, operations, function calls, and return statements. Once a variable type is established, later assignments must remain consistent with that type. For example, the following program is rejected at the type-checking stage before any execution occurs:

```
1 x = 5;
2 x = "hello";    (* TypeError: Cannot assign String to variable 'x' of
   type Integer *)
```

This contrasts with dynamically-typed languages, where the same program would run without error until (or unless) a type-dependent operation was attempted.

2.3 Token Specification

Table 2.1 summarises the major token classes used by the language.

Table 2.1. Token Specification

Token Category	Example Lexemes
INT_LITERAL	0, 10, 42
FLOAT_LITERAL	3.14, 20.5
BOOL_LITERAL	true, false
STRING_LITERAL	"hello"
IDENTIFIER	variable and function names such as x, add
Arithmetic operators (integer)	+, -, *, /
Float arithmetic operators	+, -, *, / .
Comparison operators	==, !=
Assignment operator	=
Keywords	if, else, while, def, return, print
Delimiters	() { } , ;

2.4 Context-Free Grammar

The language grammar can be described by the context-free grammar $G = (V_N, V_T, P, S)$, where V_N is the set of non-terminals, V_T is the set of terminals, P is the set of production rules, and S is the start symbol program. The grammar is implemented through recursive-descent parsing functions.

2.4.1 Program Structure

```

1 program    -> statement* EOF
2 statement  -> assignment
3           | if_stmt
4           | while_stmt
5           | func_def
6           | print_stmt
7           | return_stmt
8           | expr ';'

```

2.4.2 Expressions

```

1 expr       -> additive (('==' | '!=') additive)?
2 additive   -> term (('+' | '-' | '+.' | '-.') term)*
3 term       -> factor (('*' | '/' | '*.' | '/.') factor)*
4 factor     -> '-' factor      (* integer unary negation *)
5           | '-' factor      (* float unary negation *)
6           | INT_LITERAL
7           | FLOAT_LITERAL
8           | STRING_LITERAL
9           | BOOL_LITERAL
10          | IDENTIFIER
11          | IDENTIFIER '(' args? ')',
12          | '(' expr ')',

```

The operators `+`, `-`, `*`, and `/` operate exclusively on Integer operands. The dot-suffixed operators `+. , - . , * . ,` and `/ .` operate exclusively on Float operands. This follows the Caml Light convention of using distinct operator tokens for integer and float arithmetic, eliminating the need for implicit numeric promotion. Unary negation is similarly split: `-x` negates an Integer (desugared to `0 - x`), and `- . x` negates a Float (desugared to `0 . 0 - . x`).

2.4.3 Boolean Expressions

Comparison operators `==` and `!=` are the lowest-precedence binary operators and produce Boolean values. Because they are part of the general `expr` rule, Boolean expressions may appear both as control-flow conditions and as ordinary expressions. Their inferred type is Boolean, allowing comparison results to be assigned to variables, passed as arguments, or printed:

```
1 flag = (x == y);      (* flag inferred as Boolean *)
2 print(a != b);        (* prints true/false : Boolean *)
```

2.4.4 Statements

```
1 assignment  -> IDENTIFIER '=' expr ';'
2 if_stmt     -> 'if' '(' expr ')' block ('else' block)?
3 while_stmt  -> 'while' '(' expr ')' block
4 print_stmt  -> 'print' '(' expr ')' ';'
5 return_stmt -> 'return' expr ';'
6 block       -> '{' statement* '}'
```

2.4.5 Function Definitions and Calls

```
1 func_def -> 'def' IDENTIFIER '(' params? ')' block
2 params   -> IDENTIFIER (',' IDENTIFIER)*
3 args     -> expr (',' expr)*
```

Function calls use value parameter passing. This means the argument values are evaluated before the call and copied into the function's local environment.

2.5 Operator Precedence and Associativity

Operator precedence is encoded structurally in the recursive-descent parser. Comparison is lowest, addition and subtraction are next, and multiplication and division have the highest precedence among binary operators. Arithmetic operators are left-associative.

Table 2.2. Operator Precedence (1 = lowest, 3 = highest) and Associativity

Precedence	Operators	Associativity
1 (lowest)	==, !=	non-associative (condition use only)
2	+, -, +., -.	left-associative
3 (highest)	*, /, *., /.	left-associative

CHAPTER 3

LEXICAL ANALYSIS AND SYMBOL TABLE

This chapter covers the responsibilities related to lexical and early semantic foundations of the language. It includes defining token structures and categories (such as keywords, operators, identifiers, and literals), implementing the lexer to transform source characters into tokens, and establishing the symbol table to manage variable and function storage along with their associated types. Together, these components ensure that raw source input is systematically organised into meaningful structures that can be used reliably by later stages of the compiler.

3.1 Token representation

The token system, written in `components/tokens.py`, establishes the vocabulary and structure of all lexical elements in the language. It specifies both the types of tokens that can exist and the structure each token must follow. This serves as a canonical contract between the lexer and the parser, ensuring consistency across all stages of the compiler.

Table 3.1 summarises the main categories, example lexemes, and typical `TokenType` values.

Table 3.1. Token categories and representative `TokenType` values

Category	Example lexemes	Representative <code>TokenType</code>
Literals	42, 3.14, true, "hello"	INT_LITERAL, FLOAT_LITERAL, BOOL_LITERAL, STRING_LITERAL
Identifiers	x, counter, add	IDENTIFIER
Keywords	if, else, while, def, return, print	keyword-specific token variants
Integer operators	+, -, *, /	PLUS, MINUS, STAR, SLASH
Float operators	+., -., *., /.	PLUS_DOT, MINUS_DOT, STAR_DOT, SLASH_DOT
Other operators	=, ==, !=	ASSIGN, EQUAL_EQUAL, BANG_EQUAL
Delimiters	Parentheses, braces, comma, and semicolon (see lexer delimiter tokens)	delimiter token variants
End marker	end of source input	EOF

Token types (from class `TokenType`) are defined using an Enum with `auto()` to automatically assign unique values. This avoids reliance on raw strings and ensures safer comparisons dur-

ing parsing. The token categories include literals, identifiers, keywords, operators, delimiters, and a special end-of-file (EOF) token. The EOF token is appended after all input is processed, allowing the parser to detect the end of input explicitly rather than encountering an unexpected termination.

Each token is represented as an immutable dataclass (`frozen=True`), meaning its fields cannot be modified after creation. Each token contains four key pieces of information: its kind (`token_type`), its lexeme (the exact substring from the source code), and its source position (`line`, `column`). This structured representation allows later stages to process the program in terms of syntax and errors without handling raw text directly. Immutability improves reliability by preventing accidental modification during later processing.

```
1 @dataclass(frozen=True)
2 class Token:
3     token_type: TokenType
4     lexeme: str
5     line: int
6     column: int
```

3.2 Lexer implementation

The lexer, implemented in `components/lexica.py`, is a hand-written scanner that converts raw source text into a sequence of tokens. Unlike generated lexers (for example SLY lexer mode, or Lex and Flex), this implementation uses explicit control flow (`if` and `while`) to process input.

The lexer maintains internal state including the source string, current position, start of the current token, line number, and column index. The column index is tracked per line and resets upon encountering a newline, while a separate `token_column` records the starting position of each token for accurate error reporting.

The main function, `tokenize`, iterates through the source code, repeatedly invoking `_scan_token` to identify the next token. Tokens are appended to a list, and an EOF token is added at the end.

Whitespace is skipped, while invalid input results in a `LexerError`.

```
1 def tokenize(self) -> list[Token]:
2     tokens: list[Token] = []
3     while not self._is_at_end():
4         self.start = self.current
5         self.token_column = self.column
6         token = self._scan_token()
7         if token is not None:
```

```

8         tokens.append(token)
9     tokens.append(Token(TokenType.EOF, "", self.line, self.column))
10    return tokens

```

Token recognition follows a structured approach. Single-character tokens are mapped using a lookup table. Multi-character operators such as `==` and `!=` are handled using lookahead via `_match`. String literals are processed by `_string`, which reads characters until a closing quotation mark or the end of input is reached, with error handling for unterminated strings. Numeric literals are handled by `_number`, which distinguishes integers and floating-point values based on the presence of a decimal point followed by digits. Identifiers are processed greedily, consuming alphanumeric characters and underscores, and then checked against the `KEYWORDS` map to determine whether they are reserved words or user-defined names. Notably, `true` and `false` are also entries in this map, mapping to `TokenType.BOOL_LITERAL` rather than a keyword-specific type. This means boolean literals are recognised through the same keyword-fallback mechanism as reserved words, preventing them from being scanned as user-defined identifiers.

```

1 KEYWORDS: dict[str, TokenType] = {
2     "if": TokenType.IF,
3     "else": TokenType.ELSE,
4     "while": TokenType.WHILE,
5     # ...
6     "true": TokenType.BOOL_LITERAL,
7     "false": TokenType.BOOL_LITERAL,
8 }

```

```

1 # Arithmetic operators with optional trailing '.' for float variant
2 _ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
3     "+": (TokenType.PLUS, TokenType.PLUS_DOT),
4     "-": (TokenType.MINUS, TokenType.MINUS_DOT),
5     "*": (TokenType.STAR, TokenType.STAR_DOT),
6     "/": (TokenType.SLASH, TokenType.SLASH_DOT),
7 }

```

When one of these characters is encountered, the lexer checks the next character: if it is a `.` (and the character after that is not a digit), the dot-suffixed float variant is emitted; otherwise the plain integer operator is emitted. This lookahead prevents the `.` in a float literal such as `3.14` from being consumed as part of an operator.

```

1 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
2     "=": TokenType.ASSIGN,
3     "(": TokenType.LPAREN,
4     ")": TokenType.RPAREN,
5     # ...
6 }

```


Helper functions such as `_advance`, `_peek`, and `_peek_next` manage character consumption and lookahead, while `_is_at_end` determines when input has been fully processed. Error messages include precise line and column information, aiding debugging and diagnostics.

Table 3.2 lists the main functions and data structures in `components/lexica.py` and their roles.

Table 3.2. Key functions and data in `lexica.py`

Function / data	Responsibility	Output / effect
<code>tokenize()</code>	Iterates through source, scans tokens, appends EOF	ordered token list
<code>_scan_token()</code>	Dispatches recognition based on character and lookahead	next token or skip
<code>_number()</code>	Parses numeric sequences, distinguishes int vs float	numeric literal token
<code>_string()</code>	Parses quoted text, checks termination	string token or error
<code>_identifier()</code>	Parses identifiers, resolves keyword vs user-defined	keyword or identifier token
<code>KEYWORDS</code> map	Maps reserved words to token types	token type selection
<code>SINGLE_CHAR_TOKENS</code> map	Maps single-character symbols	token type selection
<code>_advance()</code> , <code>_peek()</code> , <code>_match()</code>	Character consumption and lookahead	scanner state update

3.3 Symbol table and semantic structure

The symbol table, defined in `components/symbol_table.py`, operates at a higher level than the lexer. While the lexer processes raw text into tokens, the symbol table manages semantic information such as variable and function names, their types, and their scope.

The symbol table is implemented as a scoped structure, where each scope maintains its own dictionary of symbols and optionally references a parent scope. This enables proper handling of nested blocks and variable shadowing. Symbols are introduced through definition functions and retrieved through lookup operations, which search recursively through parent scopes if needed.

```

1 def lookup(self, name: str) -> Symbol:
2     symbol = self._symbols.get(name)
3     if symbol is not None:
4         return symbol
5     if self.parent is not None:
6         return self.parent.lookup(name)
7     raise SymbolTableError(f"Undefined symbol: {name}")

```

The type system is represented using a `LanguageType` enumeration, which includes fundamental types such as `Integer`, `Float`, `Boolean`, `String`, and `Void`. Symbols are modelled using an inheritance hierarchy: the base `Symbol` class stores a name and type, while subclasses such as `VariableSymbol` and `FunctionSymbol` include additional attributes. Variables track initialization state, while functions store parameter lists and return types. Table 3.3 summarises the main building blocks.

Table 3.3. Symbol table components and roles

Component	Stored information	Role in semantic analysis
Scoped symbol table	Symbols in current and parent scope	Supports nesting and shadowing via recursive lookup
<code>LanguageType</code> enum	<code>Integer</code> , <code>Float</code> , <code>Boolean</code> , <code>String</code> , <code>Void</code>	Defines the type system
<code>Symbol</code> (base)	Name and declared or inferred type	Base abstraction for entries
<code>VariableSymbol</code>	Variable metadata (including initialization state)	Tracks correct usage before assignment
<code>FunctionSymbol</code>	Parameters and return type	Validates calls and return consistency
<code>lookup(name)</code>	Recursive scope search	Resolves identifiers or reports undefined
<code>SymbolTableError</code>	Semantic error details	Reports duplicates, undefined symbols, invalid operations

Errors related to semantic violations are handled via `SymbolTableError`. These include duplicate declarations within the same scope, references to undefined symbols, and invalid operations such as attempting to mark a non-variable symbol as initialized.

3.4 Role of the symbol table in the pipeline

The symbol table is constructed during the type-checking phase rather than during lexical analysis. This guideline is followed in the implementation. The type checker initialises a global symbol table and populates it while traversing the AST, including defining variables and functions, managing nested scopes, and validating type correctness.

```

1 def run_pipeline(source_code: str) -> PipelineResult:
2     tokens = Lexer(source_code).tokenize()
3     tree = Parser(tokens).parse()
4     ast_text = ASTPrinter().print(tree)
5     checked_scope = TypeChecker().check(tree)
6     outputs: list[str] = []
7     Interpreter(output_callback=outputs.append).execute(tree)
8     return PipelineResult(
9         tokens=tokens,

```

```
10     ast_text=ast_text ,
11     checked_scope=checked_scope ,
12     outputs=outputs ,
13 )
```

Once type checking is complete, the populated symbol table is returned as part of the pipeline result (`checked_scope`). The checker creates child scopes for blocks and functions during analysis, but the displayed `checked_scope` table represents the global scope entries returned by the pipeline. It can then be displayed or used for further analysis.

This separation of concerns keeps lexical analysis focused on tokenization, while semantic meaning is enforced during type checking. For user-defined functions, definitions are first registered in the global scope with placeholder parameter and return types. These are inferred from the first valid call and refined through body checking; subsequent calls must match the inferred parameter types.

CHAPTER 4

RECURSIVE-DESCENT PARSING AND THE ABSTRACT SYNTAX TREE

This chapter presents the implementation of the recursive-descent parser and the associated abstract syntax tree (AST) model. Starting from lexer-produced tokens, the parser constructs a hierarchical Program AST that represents expressions, statements, and control-flow structure in a form suitable for downstream semantic analysis and execution. Implemented syntax coverage includes arithmetic expressions, assignments, if-then-else, while, function definitions, function calls, return, and built-in `print(...)` statements. Equality operators (`==`, `!=`) are parsed as ordinary lowest-precedence expressions, so Boolean results can appear in conditions, assignments, function arguments, and `print(...)` expressions.

4.1 Abstract syntax tree representation

The AST is defined as a collection of Python dataclasses representing the structural elements of the programming language. It serves as an intermediate representation between parsing and later stages such as type checking and interpretation.

A key distinction exists between tokens and AST nodes. Tokens are flat and describe individual lexical elements such as operators or identifiers. In contrast, the AST is hierarchical. For example, while a token may represent the symbol `+`, a `BinaryOp` node represents an entire addition operation, including its left and right operands as subtrees.

All AST nodes inherit from a common base class, allowing them to store positional metadata (`line`, `column`). This ensures that errors detected in later stages can still reference precise locations in the original source code.

```
1 @dataclass
2 class ASTNode:
3     """Base class for every AST node. Carries source location for
4         diagnostics."""
5     line: int = field(default=0, kw_only=True)
6     column: int = field(default=0, kw_only=True)
```

4.1.1 Expressions

Expressions represent computations or values within the program. A `Literal` node stores values known at parse time, including integers, floating-point numbers, booleans, and strings. String literals are stored without surrounding quotation marks, as these are removed during parsing.

```
1 if tok.token_type == TokenType.STRING_LITERAL:
2     self._advance()
3     return Literal(value=tok.lexeme[1:-1], line=tok.line, column=tok.
        column)
```

An `Identifier` node represents the use of a variable name within an expression, indicating evaluation rather than declaration. A `BinaryOp` node models arithmetic and comparison operations such as addition, subtraction, multiplication, division, and equality checks, storing the operator along with references to left and right operands. A `FunctionCall` node represents function invocation, consisting of the function name and a list of argument expressions.

```
1 @dataclass
2 class BinaryOp(ASTNode):
3     """An arithmetic or comparison binary operation.
4
5     op is one of: '+', '-', '*', '/', '+.', '-.', '*.', '/.', '==', '!=',
6     Example:  a + b    x == 0    a +. b
7     """
8     op: str
9     left: ASTNode
10    right: ASTNode
```

4.1.2 Statements

Statements represent actions or control flow within the program. An `Assign` node models variable assignment, pairing a variable name with a value expression, while `Print` and `Return` nodes each store a single expression and represent output and function return operations respectively. A `Block` node groups multiple statements within braces, representing a scoped execution sequence.

Control flow is handled via `If` and `While` nodes. The `If` node stores a condition, a then-block, and an optional else-block, while `While` stores a loop condition and body block. Function definitions are represented by `FunctionDef` nodes, which include the function name, a list of parameter names, and a body block. Parameter types are intentionally not assigned at parse time, as type inference is handled later by the type checker. Finally, `Program` serves as the

AST root, containing all top-level statements.

4.2 Parser implementation

The parser consumes the list of tokens generated by the lexer and produces a single Program AST node. It is implemented as a hand-written recursive-descent parser based on the project grammar. This avoids parser generators and instead relies on explicit functions for each grammar rule.

The parser maintains internal state consisting of the token list and a position index (`self._pos`) indicating the current token. Tokens are accessed via helper methods such as `_peek` and `_advance`. Parsing errors are reported using a custom `ParseError` exception with line and column information consistent with lexer diagnostics.

```
1 def parse(self) -> Program:
2     """Parse the full token stream and return the root Program node."""
3     line, col = self._peek().line, self._peek().column
4     statements: list[ASTNode] = []
5     while not self._is_at_end():
6         statements.append(self._statement())
7     return Program(statements=statements, line=line, column=col)
```

In addition to the main `parse()` entry point, the parser's statement dispatcher (`_statement`) implements a deterministic decision order over token patterns: function definition, conditional, loop, return, print, assignment, then expression statement fallback. This design is important because it prevents ambiguity between syntactically similar forms. In particular, assignments are recognized using one-token lookahead (`IDENTIFIER` followed by `=`), while identifier-led expressions that are not assignments are parsed as expression statements and must still end with a semicolon. This gives the parser full coverage of both declarative-style and expression-driven top-level statements without requiring a parser generator.

4.2.1 Statement parsing

The main parsing loop repeatedly processes statements until EOF, with the statement type determined via dispatch on the current token. Function definitions, conditionals, loops, returns, and prints each have dedicated parsing methods. Assignments are detected using one-token lookahead (`IDENTIFIER` followed by `ASSIGN`), preventing misclassification of function calls as assignments. If no specific statement pattern matches, the parser falls back to an expression statement and enforces a terminating semicolon.

```

1 # assignment: IDENTIFIER "=" expr ";"
2 if (
3     tok.token_type == TokenType.IDENTIFIER
4     and self._peek_next().token_type == TokenType.ASSIGN
5 ):
6     return self._assignment()
7 # fallback: bare expression statement expr ";"
8 node = self._expr()
9 self._expect(TokenType.SEMICOLON, "Expected ';' after expression")

```

4.2.2 Expression parsing and precedence

Expressions are parsed in layered precedence levels: `_expr` handles comparison operators first (`==`, `!=`) after parsing an additive left-hand side, then `_additive` handles `+`, `-`, `+. .`, and `-. .`; `_term` handles `*`, `/`, `*. .`, and `/. .`; and `_factor` handles literals, identifiers and function calls, and parenthesized expressions. Function calls are recognised when an identifier is followed by `(`, with argument lists parsed as comma-separated expressions. Parenthesized expressions are supported to override precedence. Unary negation (`-x`, `-5`) is also handled at the `_factor` level: a leading `-` token causes the parser to recursively parse the operand and return `BinaryOp(0 - operand)`, so the existing type rules for subtraction cover negation without any additional inference code. Float unary negation is handled similarly for `-. .`, producing `BinaryOp(0.0 -. operand)`.

```

1 def _expr(self) -> ASTNode:      # ==, !=
2     ...
3 def _additive(self) -> ASTNode:  # +, -, +. ., -. .
4     ...
5 def _term(self) -> ASTNode:      # *, /
6     ...
7 def _factor(self) -> ASTNode:    # unary -, literals, identifiers/calls,
8     ...                          grouped expr

```

Figure 4.1 shows the AST produced for the statement `z = x + y * 2;`. Because `_term` handles `*` before `_expr` handles `+`, the multiplication sub-tree is nested one level deeper, encoding the higher precedence of `*` over `+` without any explicit priority table at runtime.

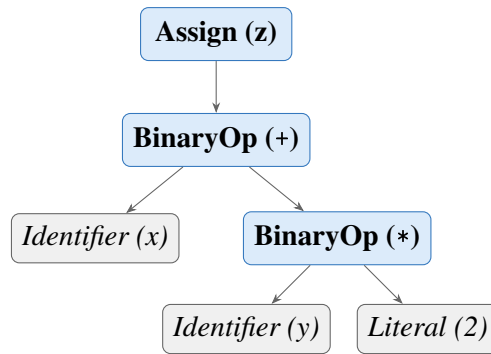


Figure 4.1. AST for `z = x + y * 2;`. Multiplication is a deeper sub-tree than addition, encoding higher precedence through tree structure rather than a runtime priority table.

This structural encoding of precedence eliminates the need for runtime operator-priority resolution and keeps semantic analysis independent of parsing strategy.

Expression parsing is intentionally split into layered methods to enforce precedence and associativity in a transparent way. Parenthesized expressions are parsed explicitly and re-enter `_expr`, allowing precedence override where required by source syntax. Comparison operators (`==`, `!=`) are parsed at the top of the `_expr` method as the lowest-precedence binary operators. Because comparisons are part of the general expression grammar, Boolean results can appear in any expression context — as conditions, assignments, arguments, or print expressions. Semantic enforcement that a condition is truly Boolean is deferred to the type checker.

Figure 4.2 shows the AST for an `if` statement with a comparison condition, illustrating how the parser structures control flow with a Boolean sub-tree.

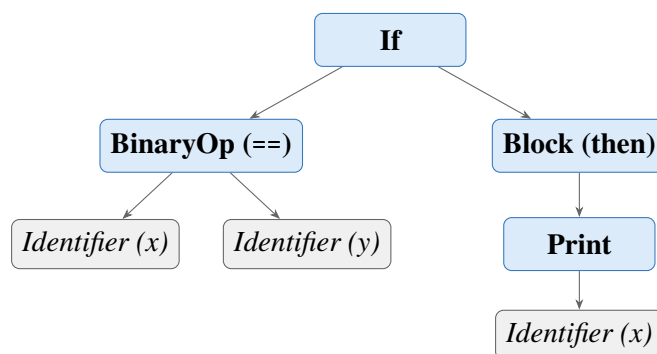


Figure 4.2. AST for `if (x == y) { print(x); }`. The comparison produces a `BinaryOp` node whose inferred type is Boolean, checked by the type checker before execution.

4.2.3 *Helper functions and lookahead*

As described above, the parser uses `_check`, `_expect`, `_peek`, `_peek_next`, `_advance`, and `_is_at_end` for token navigation and validation. Lookahead is central for ambiguity resolution: assignment versus function call is resolved by inspecting the next token, while comparison operators are recognised in `_expr` after parsing the left additive expression. Parser errors follow a consistent diagnostic format: `[line X, col Y] ParseError: <message>`.

The parser also includes dedicated list parsers for formal parameters (`_params`) and call arguments (`_args`), both implemented as comma-separated sequences with optional emptiness. This ensures consistent handling of signatures such as `def f()` and `def f(a, b)` as well as invocations like `f()` and `f(x, y+1)`.

4.3 AST printing

For debugging and reporting, an AST printer converts the tree into a readable indented string, integrated into the pipeline output. The printer uses visitor-style dynamic dispatch, mapping node class names to methods such as `_visit_If` and `_visit_BinaryOp`, with output built as multi-line text where indentation reflects tree depth. Node formatting emphasises structure: `Program` and `Block` include statement counts, `BinaryOp` explicitly shows left and right subtrees, and `FunctionCall` prints each argument by index. `Literal` includes both value and runtime Python type names for clarity.

4.4 Parser and type checker boundary

A deliberate architectural boundary is maintained between syntax construction and semantic validation. The parser is responsible for producing a structurally valid AST with accurate source locations, while type correctness and semantic constraints are checked later by the type checker. For example, `_expr` may syntactically produce a node whose eventual type is not `Boolean`, but rejection of non-`Boolean` control-flow conditions is handled during semantic analysis. This separation keeps parser logic focused on grammar conformance and allows a clear responsibility split across team member contributions.

CHAPTER 5

TYPE INFERENCE ALGORITHM

5.1 Overview

Type inference in this project means that the programmer does not write explicit type declarations for variables or functions. Instead, the compiler infers types from assignments, literals, operations, function calls, and return statements. This keeps the language simple while still preserving static type safety.

The type checker runs after parsing. It receives the AST and validates the program before execution. Any inconsistent use of types is reported as a type error, and execution is stopped.

The interpreter stage does not repeat this validation: it walks the same AST with ordinary Python representations of values. Static type information remains in the symbol table produced by the checker; run-time values are not separately labelled with `LanguageType`, except that `print()` formats output by inspecting the value's Python type.

5.2 Type Environment

The type environment is represented by the symbol table. Conceptually, it is a mapping

$$\Gamma : \text{Identifier} \rightarrow \text{Type}$$

The environment grows as the checker visits the AST. On first assignment, a variable name is inserted with its inferred type. When a new block or a function call is entered, a child environment is created so that nested scopes can be checked independently while still accessing outer bindings when needed.

5.3 Type Inference Procedure

The type checker performs type inference by traversing the abstract syntax tree before interpretation. The algorithm is syntax-directed: each AST node is visited, the types of its child nodes are inferred first, and the current node is checked according to the language typing rules.

Algorithm TYPECHECK(*node*, Γ)

Input: *node* = current AST node; Γ = current type environment / symbol table

Output: inferred type of *node*, or `TypeError`

1. **If** *node* is a **Literal**: return the corresponding primitive type: `Integer`, `Float`, `Boolean`, or `String`.
2. **If** *node* is an **Identifier**: look up the identifier in Γ . If it is not found, report an undefined variable error. Otherwise, return its stored type.
3. **If** *node* is a **BinaryOp**:
 - Infer the type of the left operand and the right operand.
 - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Integer`; return `Integer`.
 - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Float`; return `Float`.
 - If the operator is `==` or `!=`: require both operands to be the same arithmetic type; return `Boolean`.
 - Otherwise, report a `TypeError`.
4. **If** *node* is an **Assignment**:
 - Infer the type of the right-hand-side expression.
 - If the variable is not yet in Γ : insert it with the inferred type.
 - Else: verify the new type matches the existing type; if not, report a `TypeError`.
5. **If** *node* is an **If** statement:
 - Infer the condition type; require it to be `Boolean`.
 - Type-check the then-block in a child environment.
 - If an else-block exists, type-check it in a child environment.
6. **If** *node* is a **While** statement:
 - Infer the condition type; require it to be `Boolean`.
 - Type-check the loop body in a child environment.
7. **If** *node* is a **FunctionDef**:
 - Register the function name in Γ .
 - Parameter types are initially unknown.
 - Return type is inferred from `return` statements in the body.

8. If *node* is a **FunctionCall**:

- Infer all argument types.
- If this is the first valid call: store the argument types as the function’s parameter types.
- Else: compare argument types with the stored parameter types.
- Type-check the function body using a local environment.
- Return the inferred function return type.

9. If *node* is a **Print** statement: infer the expression type; accept any of the four supported types.

5.4 Inference Rules

5.4.1 Literals

Literal nodes have direct types:

$$\Gamma \vdash 42 : \text{Integer}$$
$$\Gamma \vdash 3.14 : \text{Float}$$
$$\Gamma \vdash \text{true} : \text{Boolean}$$
$$\Gamma \vdash \text{“hi”} : \text{String}$$

5.4.2 No Operator Overloading

The language does not support operator overloading. Following the Caml Light convention, integer and float arithmetic use entirely separate operator tokens, so every operator has a single, fixed type signature:

- $+$, $-$, $*$, $/$ — require two `Integer` operands; result is `Integer`.
- $+. , -. , *. , /.$ — require two `Float` operands; result is `Float`.
- $+$ is not overloaded for string concatenation; applying it to a `String` is a type error.
- $==$ and $!=$ are not overloaded for `Boolean` or `String` equality.
- No implicit numeric promotion: `Integer + Float` is a type error.

Because every operator has a unique, unambiguous type derivable from its token alone, the compiler can always infer expression types without requiring explicit type declarations from the programmer. This design ensures that operator meaning is determined uniquely from the operator token itself, eliminating semantic ambiguity during type inference and simplifying static analysis.

5.4.3 Arithmetic Expressions

The integer operators `+`, `-`, `*`, and `/` require both operands to be `Integer` and always return `Integer` (including division, which uses integer floor division). The float operators `+`, `-`, `*`, and `/` require both operands to be `Float` and always return `Float`. Mixing an `Integer` and a `Float` with any operator is a type error.

Table 5.1. Arithmetic operator type rules

Left operand	Operator	Right operand	Result type
Integer	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Integer	Integer
Float	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Float	Float
Integer	any	Float	TypeError
Float	any	Integer	TypeError
Any	any	String or Boolean	TypeError

Unary negation

Unary integer negation (`-x`) is desugared by the parser to `0 - x`. Unary float negation (`- . x`) is desugared to `0 . 0 - . x`. The type checker therefore applies the standard binary rules in both cases.

5.4.4 Boolean Expressions

The comparison operators `==` and `!=` always produce `Boolean`. Both operators require two operands of the *same* arithmetic type — either both `Integer` or both `Float`; mixed-type comparisons such as `Integer == Float` are rejected. Comparisons involving `Boolean` or `String` operands are also rejected by the type checker. Although `Boolean` and `String` values exist in the language, equality and inequality are deliberately restricted to arithmetic expressions because the project specification defines basic `Boolean` expressions as equality and inequality between two arithmetic expressions.

5.4.5 Assignment Statement

Assignments define variable types on first use. If a variable does not yet exist in the type environment, the checker infers the type of the right-hand side and inserts the variable into the environment. If the variable already exists, the new value must have the same type.

If $x \notin \Gamma$ and $\Gamma \vdash e : T$, then $\Gamma, x : T \vdash x = e$

If $x : T \in \Gamma$ and $\Gamma \vdash e : T$, then $\Gamma \vdash x = e$

5.4.6 If-Then-Else

The condition of an if statement must be Boolean. The then-block and else-block are checked in child scopes so that local operations can be validated cleanly.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B_{\text{then}} \quad \Gamma'' \vdash B_{\text{else}} \implies \Gamma \vdash \text{if}(c) B_{\text{then}} \text{ else } B_{\text{else}}$$

5.4.7 While-Loop

The condition of a while loop must also be Boolean. The loop body is checked in a child scope and must not violate the types of variables already introduced in the surrounding environment.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B \implies \Gamma \vdash \text{while}(c) B$$

5.4.8 Function Definition and Call

Function parameter types are inferred from the first call to the function. After that first call, subsequent calls must use arguments of the same types. The return type is inferred from the return statements in the function body. If the function returns inconsistent types, the checker reports an error.

For functions that are defined but never called, the type checker performs a post-pass after all top-level statements have been processed. It infers parameter types by scanning the body for arithmetic and comparison usage, then checks the full body with those inferred types. This ensures that type errors inside never-called function bodies are still detected before execution.

Recursive calls are intentionally outside the scope of this implementation; supporting them would require additional fixed-point inference or explicit recursion handling that is orthogonal to the core type-inference algorithm.

5.5 Type Error Reporting

Type errors are reported with source positions so that the user can locate the problem quickly.

The format used by the system is:

[line X, col Y] TypeError: message

Examples include assigning a String to an Integer variable, using a non-Boolean condition in an if statement, or supplying the wrong argument type to a function.

5.6 Example Type Inference Traces

Consider the following program fragment:

```
1 x = 3;
2 y = 4;
3 z = x + y;
4 print(z);
```

The checker infers:

- x: Integer (from literal 3)
- y: Integer (from literal 4)
- x + y: Integer (Integer + Integer → Integer)
- z: Integer

For a float example using dot-suffixed operators:

```
1 def addf(a, b) {
2     return a +. b;
3 }
4
5 result = addf(2.0, 3.5);
```

The first function call infers a: Float and b: Float. The operator +. requires both operands to be Float, so the function return type is inferred as Float. The variable result is also inferred as Float.

CHAPTER 6

SYSTEM ARCHITECTURE AND IMPLEMENTATION

This chapter describes how the implementation is organised: the end-to-end processing flow, repository layout, technology choices, and the main components that realise static analysis and execution (type checker, interpreter, and user interfaces). Detailed treatment of lexical analysis, the symbol table, and parsing appears in Chapters 3 and 4; Chapter 5 presents the type-inference rules that the type checker implements.

6.1 Compiler pipeline

The system follows a four-stage pipeline. Lexical and syntactic processing are kept separate from semantic analysis and execution, which simplifies testing and localises errors to the appropriate stage.

The overall flow of data through the system is shown in Figure 6.1.

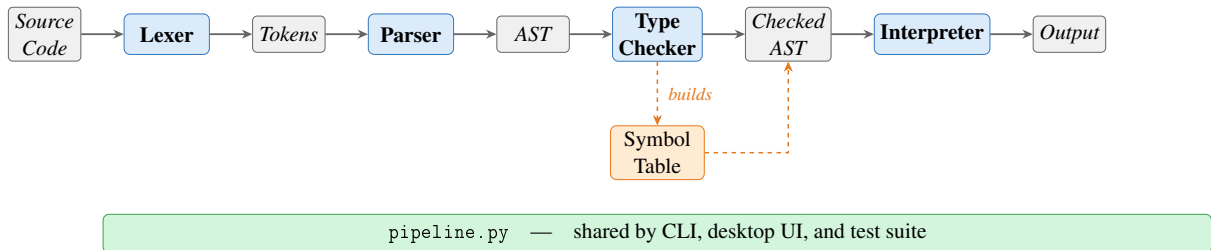


Figure 6.1. Front-end and interpreter pipeline overview.

Source characters are scanned by the lexer into a token stream. The parser consumes tokens and builds an AST that reflects program structure (expressions, statements, and control flow). The type checker then walks the AST, populates and uses the symbol table, infers types, and validates scope and type rules. Only programs that pass the type checker proceed to interpretation. The interpreter evaluates the checked tree using ordinary Python values; it does not re-run the type rules, since correctness at runtime is guaranteed by the prior static pass. All four stages are orchestrated through `pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite.

6.1.1 End-to-end pipeline walkthrough

The following example traces the complete execution of a three-statement program through all four pipeline stages, illustrating the correspondence between source text, tokens, abstract syntax tree, inferred types, and runtime output.

Source program:

```
1 a = 3;
2 b = 4;
3 print(a + b);
```

Stage 1 — Lexer output (token stream):

1	IDENTIFIER	'a'	line 1, col 1
2	ASSIGN	'='	line 1, col 3
3	INT_LITERAL	'3'	line 1, col 5
4	SEMICOLON	';'	line 1, col 6
5	IDENTIFIER	'b'	line 2, col 1
6	ASSIGN	'='	line 2, col 3
7	INT_LITERAL	'4'	line 2, col 5
8	SEMICOLON	';'	line 2, col 6
9	PRINT	'print'	line 3, col 1
10	LPAREN	'('	line 3, col 6
11	IDENTIFIER	'a'	line 3, col 7
12	PLUS	'+'	line 3, col 9
13	IDENTIFIER	'b'	line 3, col 11
14	RPAREN	)'	line 3, col 12
15	SEMICOLON	';'	line 3, col 13
16	EOF		line 3, col 14

Stage 2 — Parser output (abstract syntax tree):

```
1 Program [3 statement(s)]
2   Assign 'a' =
3     Literal 3 (int)
4   Assign 'b' =
5     Literal 4 (int)
6   Print
7     BinaryOp '+'
8       left:
9         Identifier 'a'
10      right:
11        Identifier 'b'
```

Stage 3 — Type checker output (inferred type environment):

Name	Inferred type
a	Integer
b	Integer
a + b	Integer (Integer + Integer → Integer)

Stage 4 — Interpreter output:

```
7 : Integer
```

The example demonstrates the full data flow: source characters are tokenised by the lexer, structured into an AST by the parser, validated and annotated by the type checker, and finally evaluated by the interpreter to produce typed output.

6.1.2 Entry points and shared pipeline

All stages are orchestrated by `components/pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite. In this Windows workspace, `.\.venv\Scripts\python.exe main.py` runs a script (or built-in sample) and prints the stages in order: tokens, AST text, type table, and execution output. `.\.venv\Scripts\python.exe ui.py` runs the Tkinter workbench: the same pipeline is used internally, with each stage shown in a separate tab so the user can inspect results without altering the underlying flow.

6.2 Project structure

The main source files and their responsibilities are summarised in Table 6.1.

Table 6.1. Main source files

File	Responsibility
<code>main.py</code>	Command-line runner for sample code or a script file
<code>ui.py</code>	Desktop UI for editing and running a script
<code>components/tokens.py</code>	Token classes and token categories
<code>components/lexica.py</code>	Lexical analysis
<code>components/ast_nodes.py</code>	AST node definitions
<code>components/parser.py</code>	Recursive-descent parser
<code>components/ast_printer.py</code>	AST renderer for debugging and reporting
<code>components/symbol_table.py</code>	Scoped symbol table and semantic types
<code>components/type_checker.py</code>	Static type checker and inference
<code>components/interpreter.py</code>	Tree-walking interpreter
<code>components/pipeline.py</code>	Shared pipeline for CLI, UI, and tests

6.3 Technology stack

The project uses Python and standard-library tools only (Table 6.2).

Table 6.2. Technology stack

Tool	Purpose
Python 3.10+	Core implementation language
dataclasses	AST and runtime helper structures
tkinter	Desktop graphical interface
unittest	Automated testing

6.4 Type checker

The `TypeChecker` class implements the inference and checking rules described in Chapter 5. It traverses the AST before execution, uses the symbol table to record variable and function information, infers types on first assignment, validates arithmetic and comparison expressions, enforces Boolean conditions for control flow, infers function parameter types from the first call (or from body usage for uncalled functions), and infers function return types from `return` statements. Programs that fail these rules are rejected before interpretation.

6.4.1 Scope of the implemented type system

In scope terms, this component turns a syntactically valid AST into a semantically valid program:

- static typing rules for arithmetic, comparison, assignment, control flow, and functions,
- type checking for expressions and statements,
- variable type inference from first assignment,
- function parameter inference from the first call (or from body usage for uncalled functions),
- function return type inference from `return` statements,
- source-positioned type errors when a program violates a typing rule.

6.5 Interpreter

The interpreter is a tree-walking evaluator over the checked AST. It maintains nested runtime environments for variables and function calls. Assignments write through the current or nearest enclosing scope; `if` selects a branch; `while` repeats until its condition becomes false; function calls bind arguments by value and return through an internal return mechanism.

The built-in `print()` function outputs both value and runtime type. All four types are supported:

```
1 print(42);           (* 42 : Integer      *)
2 print(3.14);         (* 3.14 : Float      *)
3 print(true);         (* true : Boolean    *)
4 print("plc");        (* "plc" : String   *)
```

6.5.1 Scope of the implemented runtime

The runtime stage provides observable behaviour and integrates with the shared pipeline:

- assignment execution,
- `if` execution,
- `while` execution,
- function execution with value parameter passing,
- `print()` execution with immediate output,
- execution behind the same `run_pipeline` path used by the CLI, UI, and tests.

6.6 Graphical user interface

The desktop workbench (`ui.py`) is implemented with Python's `tkinter` standard library and requires no additional dependencies. The window is titled *125970-126690-126687-Ass2-PLC* and opens at 1200×760 pixels in a horizontal split layout.

Toolbar. A top toolbar provides three buttons — *Open Script*, *Run*, and *Clear Output* — and a right-aligned status label that reports *Ready*, a loaded filename, or *Run failed*.

Left panel (source editor). A monospaced Text widget with both horizontal and vertical scrollbars allows the user to enter or paste arbitrary source code. The *Open Script* button

populates the editor from a file. The editor is pre-loaded with the built-in sample program on first launch.

Right panel (output tabs). A Notebook widget presents four tabs, each backed by a scrollable read-only Text widget with both horizontal and vertical scrollbars:

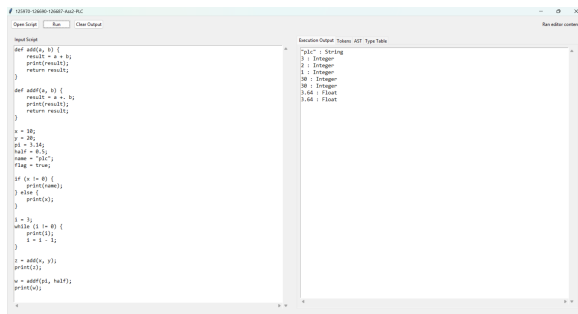
- *Execution Output* — the `print()` values produced by stage 4, formatted as `value : Type`. Runtime output includes both the computed value and its inferred static type, reflecting the result of the type-checking phase before interpretation;
- *Tokens* — the token stream from stage 1, one token per line with type, lexeme, and source position;
- *AST* — the indented tree text from stage 2;
- *Type Table* — the symbol table from stage 3 showing each name, kind, inferred type, initialisation state, and parameter list.

Error handling. If any pipeline stage raises an exception, the error message is written to the *Execution Output* tab as `ERROR: <message>`. No modal dialogs are used, so the user can edit and rerun without dismissing a popup.

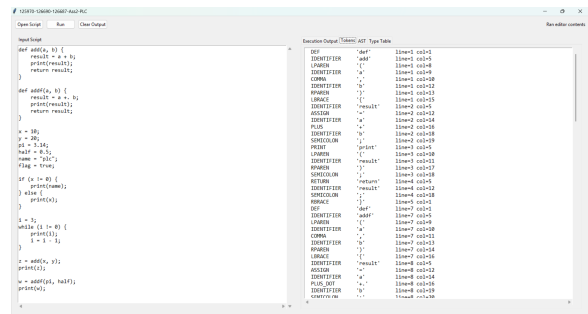
Keyboard shortcuts. F5 and Ctrl+Enter are both bound to the *Run* action, allowing rapid edit-run cycles without reaching for the mouse.

6.6.1 Screenshots

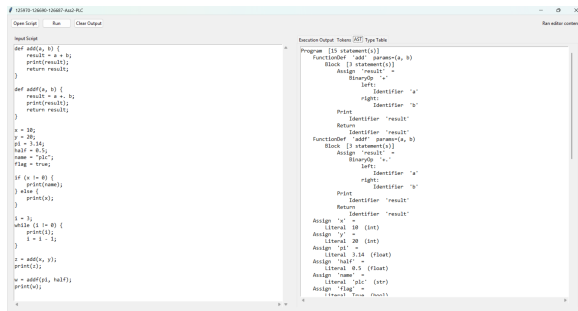
Figures 6.2 and 6.3 show the desktop workbench and CLI output.



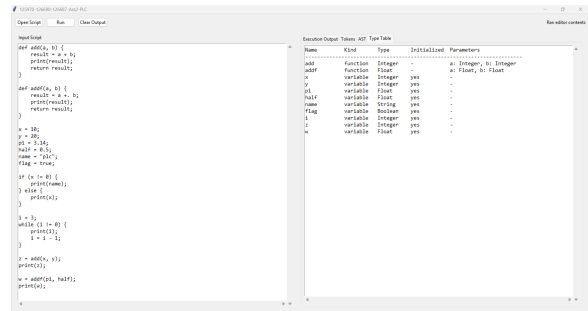
(a) Execution Output tab.



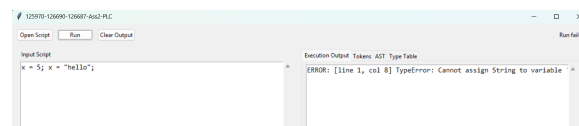
(b) Tokens tab (Stage 1).



(c) AST tab (Stage 2).



(d) Type Table tab (Stage 3).



(e) Inline type-error display.

Figure 6.2. Desktop workbench — output tabs and error display.

6.7 Command-line interface

The CLI runner (`main.py`) accepts an optional script path; without one it runs the built-in sample and prints four labelled stages to `stdout`.

```

PS C:\Users\windev\Desktop\125970-126690-126687-Assignment2-PLC [set-externalenv] > .\ps C:\Users\windev\Desktop\125970-126690-126687-Assignment2-PLC\venv\Scripts\Activate.ps1
(.venv) PS C:\Users\windev\Desktop\125970-126690-126687-Assignment2-PLC> python main.py

=====
STAGE 1 - TOKENS
=====
def      'def'      line=2 col=1
IDENTIFIER 'add'      line=2 col=5
LPAREN   '('        line=2 col=8
IDENTIFIER 'a'       line=2 col=9
COMMA    ','        line=2 col=10
IDENTIFIER 'b'       line=2 col=12
RPAREN   ')'        line=2 col=13
LBRACE   '{'        line=2 col=15
IDENTIFIER 'result'  line=2 col=16
ASSIGN   '='        line=3 col=12
IDENTIFIER 'a'       line=3 col=14
PLUS     '+'        line=3 col=16
IDENTIFIER 'b'       line=3 col=18
SEMICOLON ';'        line=3 col=19
PRINT   'print'     line=4 col=5
LPAREN   '('        line=4 col=8
IDENTIFIER 'result'  line=4 col=11
RPAREN   ')'        line=4 col=13
SEMICOLON ';'        line=4 col=14
RETURN  'return'    line=5 col=5
IDENTIFIER 'result'  line=5 col=12
SEMICOLON ';'        line=5 col=14
Z       'z'         line=5 col=16
RBRACE   '}'        line=6 col=1
DEF      'def'      line=6 col=1
IDENTIFIER 'add'      line=6 col=5
LPAREN   '('        line=6 col=8
IDENTIFIER 'a'       line=6 col=9
COMMA    ','        line=6 col=10
IDENTIFIER 'b'       line=6 col=12
RPAREN   ')'        line=6 col=14
LBRACE   '{'        line=6 col=16
IDENTIFIER 'result'  line=6 col=17
ASSIGN   '='        line=6 col=18
IDENTIFIER 'a'       line=6 col=20
SEMICOLON ';'        line=6 col=21

```

(a) Stage 1: token stream.

```

Assign 'w' =
FunctionCall 'addf' [2 arg(s)]
arg[0]:
Identifier 'pi'
arg[1]:
Identifier 'half'

Print
Identifier 'w'

=====
STAGE 3 - TYPE CHECK
=====
Name      Kind      Type      Initialized Parameters
-----
add        function  Integer   -          a: Integer, b: Integer
addf       function  Float     -          a: Float, b: Float
x          variable  Integer   yes         -
y          variable  Integer   yes         -
pi         variable  Float     yes         -
half       variable  Float     yes         -
name       variable  String    yes         -
flag       variable  Boolean   yes         -
i          variable  Integer   yes         -
z          variable  Integer   yes         -
w          variable  Float     yes         -

=====
STAGE 4 - EXECUTION
=====
"plc" : String
3 : Integer
2 : Integer
1 : Integer
30 : Integer
30 : Integer
3.64 : Float
3.64 : Float
(.venv)[ps C:\Users\windev\Desktop\125970-126690-126687-Assignment2-PLC>

```

(b) Stage 3 type table and Stage 4 output.

Figure 6.3. CLI run of python main.py on the built-in sample.

CHAPTER 7

TESTING AND VERIFICATION

7.1 Manual Testing

Prior to the construction of the automated suite, each language feature was exercised manually by running the full pipeline and inspecting stage outputs. Table 7.1 summarises the categories covered.

Table 7.1. Representative manually tested categories

Category	Program Behaviour	Expected Result
Arithmetic	Evaluate integer and float expressions separately	Correct value and inferred type
If-then-else	Choose one branch from a Boolean condition	Only matching branch runs
While-loop	Print during repeated execution	One output line per iteration
Functions	Infer parameters and return type	Valid call and correct return value
Type mismatch	Reassign incompatible type	Type error before runtime

7.1.1 *Lexer and Symbol Table*

The lexer was verified manually by running the full pipeline and inspecting Stage 1 token output for programs that exercise every token category: integer and float literals, Boolean literals, string literals, all six keywords, two-character comparison operators (`==`, `!=`), single-character operators, and delimiters. Identifier scanning with keyword fallback was confirmed by checking that `true`, `false`, and reserved words are emitted with their keyword token types while non-keyword names produce `IDENTIFIER` tokens. For example, the source fragment `x = 10;` produces the token sequence:

```

1 IDENTIFIER  'x'      line=1 col=1
2 ASSIGN      '='      line=1 col=3
3 INT_LITERAL '10'     line=1 col=5
4 SEMICOLON   ';'      line=1 col=7

```

Error paths were also checked: supplying a bare `!` character raises a `LexerError` with a source

position and a message suggesting !=, and an unterminated string literal raises a `LexerError` citing the opening line.

The symbol table was verified through the type checker and the Stage 3 output. Duplicate variable definitions were confirmed to raise a `SymbolTableError` before execution. Scope isolation was verified with nested blocks: a variable defined inside an `if` or `while` body is not visible in the outer scope after the block ends. The `format_table()` output was checked against the expected column layout (name, kind, type, initialisation status, parameters) for both variable and function symbols.

7.1.2 *Parser and AST*

The parser was verified manually by running the full pipeline and inspecting Stage 2 AST output. The following cases were checked:

- arithmetic expressions with mixed precedence (e.g. `x + y * 2`) produce a tree where multiplication is a deeper sub-tree than addition, confirming the `_expr/_term/_factor` precedence encoding,
- assignment versus expression statement dispatch: `x = 5`; produces an `Assign` node while `add(1,2)`; produces a `FunctionCall` node at the statement level, confirming the IDENTIFIER+ASSIGN lookahead,
- `if` with and without an `else` clause: the resulting `If` node has a non-null `else_block` only when the clause is present,
- function definitions store only parameter names in the `FunctionDef` node; the AST printer output was inspected to confirm no type annotations are present at this stage,
- string literals in `Literal` nodes have their surrounding double-quotes stripped; `print("hello")` prints `"hello"` : `String` at runtime (the interpreter re-adds the quotes when formatting string output),
- parse errors: missing semicolons, unmatched parentheses, and unexpected tokens all produce `[line X, col Y] ParseError:` messages with correct source positions.

7.1.3 Arithmetic Expressions

Arithmetic expressions were tested with integer-only and float-only programs to verify operator precedence and strict type separation. The expression `x + y * 2` confirmed that multiplication binds more tightly than addition, and mixing integer and float operands correctly produced a type error, confirming the no-overloading design. The test suite explicitly verifies that mixed-type arithmetic is rejected:

- `1 + 2` — valid Integer arithmetic.
- `2.0 + . 3.5` — valid Float arithmetic.
- `1 + 2.5` — `TypeError: operator + requires two Integer operands.`
- `2 + . 3` — `TypeError: operator + . requires two Float operands.`

7.1.4 Boolean Expressions

Boolean expressions were tested inside `if` and `while` conditions. The type checker correctly enforced that both operands of `==` and `!=` must be arithmetic; supplying a `Boolean` or `String` operand produced a source-positioned `TypeCheckError` before any execution.

7.1.5 Assignment Statements

Assignments were tested for both first-use type inference and reassignment consistency. Assigning `5` to a variable and subsequently assigning `"hello"` to the same variable correctly produced a type error, confirming that the inferred type is fixed after first assignment.

7.1.6 If-Then-Else

Conditional programs were constructed to verify that exactly one branch executes for a given condition and that neither branch is entered when the condition evaluates to the opposite truth value. Non-Boolean conditions were also supplied and confirmed to be rejected by the type checker before execution.

7.1.7 While-Loop

While-loops were tested with an integer countdown from three to zero. The results confirmed that the loop body executes once per iteration and terminates when the condition becomes false, with `print()` producing one output line per pass rather than accumulating output until loop exit.

7.1.8 Functions

Function tests covered value parameter passing, local scope execution, and return value propagation. The type checker correctly inferred parameter types from the first call site and enforced that signature for all subsequent calls to the same function.

7.1.9 Print

The built-in `print()` function was tested with all supported runtime types. The output includes the printed value together with its type, such as `3.14 : Float` and `"plc" : String`.

7.1.10 Unary Minus

The unary minus operator was verified for integer literals (`x = -5`), float literals (`y = -3.14`), negated variable references (`y = -x`), and negation inside a larger expression (`x = 3 + -2`). In each case the type checker assigned the correct type and the interpreter produced the expected value.

7.2 Type Error Detection

The checker was verified with invalid programs involving incompatible assignment, invalid arithmetic operands, wrong function arguments, and non-Boolean conditions. In each case the system stopped before execution and reported a source-positioned type error. Representative error messages are shown below.

```
1 # Assignment type mismatch
2 [line 2, col 1] TypeError: Cannot assign String to variable 'x' of type
   Integer
3
4 # Non-Boolean if condition
5 [line 1, col 4] TypeError: If condition must have type Boolean
6
7 # Wrong argument type for function
8 [line 5, col 9] TypeError: Argument 1 of 'add' must be Integer, got
   String
```

7.3 Automated Test Suite

The automated regression suite is located in `tests/test_pipeline.py` and uses Python's built-in `unittest` framework. The tests are organised into five classes, each targeting a distinct compiler stage.

Table 7.2. Automated test coverage by class

Test class	Stage covered	Tests
LexerTests	Tokenisation (<code>lexica.py</code>)	12
SymbolTableTests	Scoped symbol table (<code>symbol_table.py</code>)	7
ParserTests	Parsing and precedence (<code>parser.py</code>)	5
TypeCheckerTests	Static type inference (<code>type_checker.py</code>)	18
IntegrationTests	Full pipeline execution	20
Total		62

LexerTests verify that each token category (integer, float, Boolean, string, keywords, identifiers, two-character operators) is produced correctly, that source line numbers are tracked across newlines, and that illegal characters and unterminated strings raise `LexerError`.

SymbolTableTests exercise the symbol table directly, without going through the full pipeline. They cover variable definition and lookup, duplicate definition raising `SymbolTableError` for both variables and functions, parent-scope lookup through `child_scope()`, undefined-symbol lookup error, and the `format_table()` output for both variable and function symbol rows.

TypeCheckerTests verify static type inference and error detection. The 18 tests cover integer and float arithmetic (including dot-suffixed operators), integer floor division, strict type separation (mixed Integer/Float rejected for both arithmetic and comparison operators), string and boolean type inference, assignment type mismatch, non-Boolean conditions, undefined variables, wrong argument count and type, and two tests for uncalled-function body checking: one confirms that a type error inside a never-called function body is detected, and another confirms that a valid uncalled function raises no error.

IntegrationTests run programs through all four pipeline stages and verify the final output. Cases include unary negation on all numeric types, both branches of `if/else`, while-loop countdown, `print()` with every type, value-parameter isolation, nested function calls, variable reassignment, and scope isolation (variables declared inside an `if` block are not visible outside it).

The tests can be run with:

```
1 .\venv\Scripts\python.exe -m unittest discover -s tests -v
```

All 62 tests pass.

7.4 Error Handling

The language system reports errors at three main stages:

- lexical errors for malformed characters or unterminated strings,
- parse errors for invalid syntax such as missing semicolons,
- type errors for programs that violate static typing rules.

Runtime errors are also reported with source positions whenever execution reaches an invalid state such as an undefined variable reference.

CHAPTER 8

CONCLUSION

This project delivered a complete, statically-typed interpreted language supporting Integer, Float, Boolean, and String types, with arithmetic, comparisons, assignments, if-then-else, while-loops, functions with value parameter passing, and a built-in `print()` function.

The system follows a four-stage pipeline: lexer, parser, type checker, and interpreter. Variable and function types are inferred without explicit declarations; programs that violate type rules are rejected before execution. For uncalled functions, the type checker performs a post-pass to infer parameter types from body usage, ensuring type errors are always caught.

The 62 automated tests across five classes — lexical analysis, symbol table, parsing, type checking, and full pipeline — confirm correct behaviour for all language features. Both a command-line runner and a desktop UI are provided.

The project demonstrates how lexical analysis, parsing, semantic analysis, type inference, and interpretation can be integrated into a coherent compiler pipeline with clear separation of responsibilities between stages. The implementation prioritised semantic clarity, deterministic type inference, and modular compiler structure over advanced language features. Future work could extend the language with recursive functions, richer data structures, and code generation.

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APPENDIX A: SOURCE CODE

The complete source code is available on GitHub:

<https://github.com/The-Token-Trio/125970-126690-126687-Assignment2-PLC>

The project can be run from the command line or the desktop UI:

```
1 .\venv\Scripts\python.exe main.py
2 .\venv\Scripts\python.exe ui.py
3 .\venv\Scripts\python.exe -m unittest discover -s tests -v
```

No external Python dependencies are required.

APPENDIX B: TEAM MEMBERS AND CONTRIBUTIONS

Name	Student ID	Contributions
Aye Khin Khin Hpone (Yolanda Lim)	st125970	Static typing rules; type checking; assignment, if, while, function, and <code>print()</code> execution; unary minus inference; pipeline integration (<code>pipeline.py</code>); automated test suite (62 tests)
Applegate T. Tun Oo	st126690	Lexer implementation (scanning, tokenisation, line/column tracking, error reporting); token definitions; keywords, operators, identifiers, and literals; symbol table (variable/function/type storage)
Win Htut Naing	st126687	Arithmetic and boolean expressions; assignment, if-then-else, while-loop, function definition/call, and <code>print()</code> parsing; unary minus parsing

A.1 Entry Points

```
1 from __future__ import annotations
2
3 import argparse
4 from pathlib import Path
5
6 from components.pipeline import format_stage_output, run_pipeline
7
8
```



```

9  DEFAULT_SOURCE = """
10 def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 def addf(a, b) {
17     result = a +. b;
18     print(result);
19     return result;
20 }
21
22 x = 10;
23 y = 20;
24 pi = 3.14;
25 half = 0.5;
26 name = "plc";
27 flag = true;
28
29 if (x != 0) {
30     print(name);
31 } else {
32     print(x);
33 }
34
35 i = 3;
36 while (i != 0) {
37     print(i);
38     i = i - 1;
39 }
40
41 z = add(x, y);
42 print(z);
43
44 w = addf(pi, half);
45 print(w);
46 """
47
48
49 def run_source(source_code: str) -> None:
50     result = run_pipeline(source_code)
51     print(format_stage_output(result))
52
53
54 def _parse_args() -> argparse.Namespace:
55     parser = argparse.ArgumentParser(description="Run the programming
56         language pipeline.")
57     parser.add_argument("script", nargs="?", help="Optional path to a
58         source file to run")
59     return parser.parse_args()

```

```

60 def main() -> None:
61     args = _parse_args()
62     if args.script:
63         source_code = Path(args.script).read_text(encoding="utf-8")
64     else:
65         source_code = DEFAULT_SOURCE
66     run_source(source_code)
67
68
69 if __name__ == "__main__":
70     main()

```

Listing 8.1: main.py — command-line entry point

```

1  from __future__ import annotations
2
3  import tkinter as tk
4  from pathlib import Path
5  from tkinter import filedialog, ttk
6
7  from components.pipeline import format_tokens, run_pipeline
8
9
10 DEFAULT_SOURCE = """def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 def addf(a, b) {
17     result = a +. b;
18     print(result);
19     return result;
20 }
21
22 x = 10;
23 y = 20;
24 pi = 3.14;
25 half = 0.5;
26 name = \"plc\";
27 flag = true;
28
29 if (x != 0) {
30     print(name);
31 } else {
32     print(x);
33 }
34
35 i = 3;
36 while (i != 0) {
37     print(i);
38     i = i - 1;
39 }

```

```

40
41 z = add(x, y);
42 print(z);
43
44 w = addf(pi, half);
45 print(w);
46 """
47
48
49 class LanguageWorkbench:
50     def __init__(self) -> None:
51         self.root = tk.Tk()
52         self.root.title("125970-126690-126687-Ass2-PLC")
53         self.root.geometry("1200x760")
54         self.current_file: Path | None = None
55
56         self._build_layout()
57         self.source_text.insert("1.0", DEFAULT_SOURCE)
58         self.root.bind("<F5>", lambda _e: self._run_source())
59         self.root.bind("<Control-Return>", lambda _e: self._run_source
60             ())
61
62     def _build_layout(self) -> None:
63         toolbar = ttk.Frame(self.root, padding=10)
64         toolbar.pack(fill=tk.X)
65
66         ttk.Button(toolbar, text="Open Script", command=self.
67             _open_script).pack(side=tk.LEFT)
68         ttk.Button(toolbar, text="Run", command=self._run_source).pack(
69             side=tk.LEFT, padx=(8, 0))
70         ttk.Button(toolbar, text="Clear Output", command=self.
71             _clear_output).pack(side=tk.LEFT, padx=(8, 0))
72
73         self.status_var = tk.StringVar(value="Ready")
74         ttk.Label(toolbar, textvariable=self.status_var).pack(side=tk.
75             RIGHT)
76
77         body = ttk.Panedwindow(self.root, orient=tk.HORIZONTAL)
78         body.pack(fill=tk.BOTH, expand=True, padx=10, pady=(0, 10))
79
80         left_panel = ttk.Frame(body, padding=8)
81         right_panel = ttk.Frame(body, padding=8)
82         body.add(left_panel, weight=1)
83         body.add(right_panel, weight=1)
84
85         ttk.Label(left_panel, text="Input Script").pack(anchor=tk.W)
86         src_frame = ttk.Frame(left_panel)
87         src_frame.pack(fill=tk.BOTH, expand=True, pady=(6, 0))
88         self.source_text = tk.Text(src_frame, wrap=tk.NONE, font=(
89             "Consolas", 11))
90         src_vsb = ttk.Scrollbar(src_frame, orient=tk.VERTICAL, command=
91             self.source_text.yview)
92         src_hsb = ttk.Scrollbar(src_frame, orient=tk.HORIZONTAL,

```

```

        command=self.source_text.xview)
86 self.source_text.configure(yscrollcommand=src_vsb.set,
        xscrollcommand=src_hsb.set)
87 src_vsb.pack(side=tk.RIGHT, fill=tk.Y)
88 src_hsb.pack(side=tk.BOTTOM, fill=tk.X)
89 self.source_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
90
91 notebook = ttk.Notebook(right_panel)
92 notebook.pack(fill=tk.BOTH, expand=True)
93
94 self.output_text = self._make_output_tab(notebook, "Execution
        Output")
95 self.tokens_text = self._make_output_tab(notebook, "Tokens")
96 self.ast_text = self._make_output_tab(notebook, "AST")
97 self.types_text = self._make_output_tab(notebook, "Type Table")
98
99 def _make_output_tab(self, notebook: ttk.Notebook, title: str) ->
    tk.Text:
100     frame = ttk.Frame(notebook, padding=8)
101     notebook.add(frame, text=title)
102     text = tk.Text(frame, wrap=tk.NONE, font=("Consolas", 11),
        state=tk.DISABLED)
103     vsb = ttk.Scrollbar(frame, orient=tk.VERTICAL, command=text.
        yview)
104     hsb = ttk.Scrollbar(frame, orient=tk.HORIZONTAL, command=text.
        xview)
105     text.configure(yscrollcommand=vsb.set, xscrollcommand=hsb.set)
106     vsb.pack(side=tk.RIGHT, fill=tk.Y)
107     hsb.pack(side=tk.BOTTOM, fill=tk.X)
108     text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
109     return text
110
111 def _open_script(self) -> None:
112     file_path = filedialog.askopenfilename(
113         title="Open Script File",
114         filetypes=[("Text Files", "*.txt *.pl *.src"), ("All Files"
            , ".*")],
115     )
116     if not file_path:
117         return
118
119     path = Path(file_path)
120     self.current_file = path
121     self.source_text.delete("1.0", tk.END)
122     self.source_text.insert("1.0", path.read_text(encoding="utf-8")
        )
123     self.status_var.set(f"Loaded {path.name}")
124
125 def _run_source(self) -> None:
126     source = self.source_text.get("1.0", tk.END)
127     try:
128         result = run_pipeline(source)
129     except Exception as error:

```

```

130         self._write_text(self.output_text, f"ERROR: {error}")
131         self.status_var.set("Run failed")
132         return
133
134     self._write_text(self.output_text, "\n".join(result.outputs) if
135                     result.outputs else "(no output)")
136     self._write_text(self.tokens_text, format_tokens(result.tokens)
137                     )
138     self._write_text(self.ast_text, result.ast_text)
139     self._write_text(self.types_text, result.checked_scope.
140                     format_table())
141
142     current_label = self.current_file.name if self.current_file is
143                     not None else "editor contents"
144     self.status_var.set(f"Ran {current_label}")
145
146     def _clear_output(self) -> None:
147         self._write_text(self.output_text, "")
148         self._write_text(self.tokens_text, "")
149         self._write_text(self.ast_text, "")
150         self._write_text(self.types_text, "")
151         self.status_var.set("Output cleared")
152
153     @staticmethod
154     def _write_text(widget: tk.Text, content: str) -> None:
155         widget.config(state=tk.NORMAL)
156         widget.delete("1.0", tk.END)
157         widget.insert("1.0", content)
158         widget.config(state=tk.DISABLED)
159
160     def run(self) -> None:
161         self.root.mainloop()
162
163 if __name__ == "__main__":
164     LanguageWorkbench().run()

```

Listing 8.2: ui.py — Tkinter desktop workbench

A.2 Core Components

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4 from enum import Enum, auto
5
6
7 class TokenType(Enum):
8     # Literals
9     INT_LITERAL = auto()
10    FLOAT_LITERAL = auto()
11    BOOL_LITERAL = auto()

```

```

12  STRING_LITERAL = auto()
13
14  # Identifier
15  IDENTIFIER = auto()
16
17  # Keywords
18  IF = auto()
19  ELSE = auto()
20  WHILE = auto()
21  DEF = auto()
22  RETURN = auto()
23  PRINT = auto()
24
25  # Operators
26  PLUS = auto()
27  MINUS = auto()
28  STAR = auto()
29  SLASH = auto()
30  PLUS_DOT = auto()    # +. (float addition)
31  MINUS_DOT = auto()   # -. (float subtraction)
32  STAR_DOT = auto()    # *. (float multiplication)
33  SLASH_DOT = auto()   # /. (float division)
34  EQUAL_EQUAL = auto()
35  BANG_EQUAL = auto()
36  ASSIGN = auto()
37
38  # Delimiters
39  LPAREN = auto()
40  RPAREN = auto()
41  LBRACE = auto()
42  RBRACE = auto()
43  COMMA = auto()
44  SEMICOLON = auto()
45
46  EOF = auto()
47
48
49  @dataclass(frozen=True)
50  class Token:
51      token_type: TokenType
52      lexeme: str
53      line: int
54      column: int

```

Listing 8.3: components/tokens.py — token type enumeration and Token dataclass

```

1  from __future__ import annotations
2
3  from typing import Iterator
4
5  from components.tokens import Token, TokenType
6
7

```

```

8 KEYWORDS: dict[str, TokenType] = {
9     "if": TokenType.IF,
10    "else": TokenType.ELSE,
11    "while": TokenType.WHILE,
12    "def": TokenType.DEF,
13    "return": TokenType.RETURN,
14    "print": TokenType.PRINT,
15    "true": TokenType.BOOL_LITERAL,
16    "false": TokenType.BOOL_LITERAL,
17 }
18
19
20 # Arithmetic operators that may have a trailing '.' for the float
21 variant
22 _ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
23     "+": (TokenType.PLUS, TokenType.PLUS_DOT),
24     "-": (TokenType.MINUS, TokenType.MINUS_DOT),
25     "*": (TokenType.STAR, TokenType.STAR_DOT),
26     "/": (TokenType.SLASH, TokenType.SLASH_DOT),
27 }
28
29 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
30     "=": TokenType.ASSIGN,
31     "(": TokenType.LPAREN,
32     ")": TokenType.RPAREN,
33     "{": TokenType.LBRACE,
34     "}": TokenType.RBRACE,
35     ",": TokenType.COMMA,
36     ";": TokenType.SEMICOLON,
37 }
38
39 class LexerError(Exception):
40     pass
41
42
43 class Lexer:
44     def __init__(self, source: str) -> None:
45         self.source = source
46         self.start = 0
47         self.current = 0
48         self.line = 1
49         self.column = 1
50         self.token_column = 1
51
52     def tokenize(self) -> list[Token]:
53         tokens: list[Token] = []
54         while not self._is_at_end():
55             self.start = self.current
56             self.token_column = self.column
57             token = self._scan_token()
58             if token is not None:
59                 tokens.append(token)

```

```

60     tokens.append(Token(TokenType.EOF, "", self.line, self.column))
61     return tokens
62
63     def _scan_token(self) -> Token | None:
64         char = self._advance()
65
66         # Whitespace handling
67         if char in {" ", "\t", "\r"}:
68             return None
69         if char == "\n":
70             self.line += 1
71             self.column = 1
72             return None
73
74         # Two-char operators
75         if char == "=":
76             if self._match("="):
77                 return self._make_token(TokenType.EQUAL_EQUAL)
78             return self._make_token(TokenType.ASSIGN)
79         if char == "!":
80             if self._match("="):
81                 return self._make_token(TokenType.BANG_EQUAL)
82             raise LexerError(self._error_message("Unexpected '!' . Did
              you mean '!='?"))
83
84         # Arithmetic operators: + - * / (with optional trailing '.'
              for float variant)
85         arith = _ARITH_DOT.get(char)
86         if arith is not None:
87             int_tok, float_tok = arith
88             # Unary-minus lookahead: '-' followed by a digit is NOT
              '-' even if
89             # next char is '.' -- we still need to check char after '.'
              is not digit
90             # (to avoid consuming the '.' of a float literal like 3.14)
91             # Rule: X. is a float operator only when the char after '.'
              is NOT a digit.
92             if self._peek() == "." and not self._peek_next().isdigit():
93                 self._advance() # consume '.'
94                 return self._make_token(float_tok)
95             return self._make_token(int_tok)
96
97         # Single-char operators/delimiters
98         single_char_token = SINGLE_CHAR_TOKENS.get(char)
99         if single_char_token is not None:
100             return self._make_token(single_char_token)
101
102         if char == '"':
103             return self._string()
104
105         if char.isdigit():
106             return self._number()

```



```

107
108     if self._is_identifier_start(char):
109         return self._identifier()
110
111     raise LexerError(self._error_message(f"Unexpected character: {
112         char!r}"))
113
114 def _string(self) -> Token:
115     while self._peek() != '"' and not self._is_at_end():
116         if self._peek() == "\n":
117             self.line += 1
118             self.column = 1
119             self._advance()
120
121     if self._is_at_end():
122         raise LexerError(self._error_message("Unterminated string
123             literal"))
124
125     # Closing quote
126     self._advance()
127     return self._make_token(TokenType.STRING_LITERAL)
128
129 def _number(self) -> Token:
130     while self._peek().isdigit():
131         self._advance()
132
133     is_float = False
134     if self._peek() == "." and self._peek_next().isdigit():
135         is_float = True
136         self._advance()
137         while self._peek().isdigit():
138             self._advance()
139
140     if is_float:
141         return self._make_token(TokenType.FLOAT_LITERAL)
142     return self._make_token(TokenType.INT_LITERAL)
143
144 def _identifier(self) -> Token:
145     while self._is_identifier_part(self._peek()):
146         self._advance()
147
148     lexeme = self._current_lexeme()
149     token_type = KEYWORDS.get(lexeme, TokenType.IDENTIFIER)
150     return self._make_token(token_type)
151
152 def _current_lexeme(self) -> str:
153     return self.source[self.start : self.current]
154
155 def _make_token(self, token_type: TokenType) -> Token:
156     return Token(token_type, self._current_lexeme(), self.line,
157         self.token_column)
158
159 def _advance(self) -> str:

```

```

157     char = self.source[self.current]
158     self.current += 1
159     self.column += 1
160     return char
161
162     def _match(self, expected: str) -> bool:
163         if self._is_at_end():
164             return False
165         if self.source[self.current] != expected:
166             return False
167
168         self.current += 1
169         self.column += 1
170         return True
171
172     def _peek(self) -> str:
173         if self._is_at_end():
174             return "\0"
175         return self.source[self.current]
176
177     def _peek_next(self) -> str:
178         if self.current + 1 >= len(self.source):
179             return "\0"
180         return self.source[self.current + 1]
181
182     def _is_at_end(self) -> bool:
183         return self.current >= len(self.source)
184
185     @staticmethod
186     def _is_identifier_start(char: str) -> bool:
187         return char.isalpha() or char == "_"
188
189     @staticmethod
190     def _is_identifier_part(char: str) -> bool:
191         return char.isalnum() or char == "_"
192
193     def _error_message(self, message: str) -> str:
194         return f"[line {self.line}, col {self.token_column}] {message}"
195
196
197     def iter_tokens(source: str) -> Iterator[Token]:
198         yield from Lexer(source).tokenize()

```

Listing 8.4: components/lexica.py — lexical analyser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from typing import Optional
5
6
7 #

```

```

8  # Base
9  #
10
11 @dataclass
12 class ASTNode:
13     """Base class for every AST node. Carries source location for
14         diagnostics."""
15     line: int = field(default=0, kw_only=True)
16     column: int = field(default=0, kw_only=True)
17
18 #
19
20 # Expressions
21 #
22
23 @dataclass
24 class Literal(ASTNode):
25     """An integer, float, boolean, or string literal.
26
27     Examples:  42    3.14    true    "hello"
28
29     value: int | float | bool | str
30
31 @dataclass
32 class Identifier(ASTNode):
33     """A variable or function name used as an expression.
34
35     Example:  x
36
37     name: str
38
39 @dataclass
40 class BinaryOp(ASTNode):
41     """An arithmetic or comparison binary operation.
42
43     op is one of: '+' '-' '*' '/' '+.' '-.' '*.' '/.' '==' '!='
44     Example:  a + b    x == 0    a +. b
45
46     op: str
47     left: ASTNode
48     right: ASTNode
49
50 @dataclass
51
52

```

```

53 class FunctionCall(ASTNode):
54     """A call to a user-defined function.
55
56     Example:  add(1, 2)
57     """
58     name: str
59     args: list[ASTNode] = field(default_factory=list)
60
61
62 #
63 # -----
64 #
65 # Statements
66 # -----
67
68 @dataclass
69 class Assign(ASTNode):
70     """Variable assignment (first use infers type; later uses must
71     match).
72
73     Example:  x = 10;
74     """
75     name: str
76     value: ASTNode
77
78 @dataclass
79 class Print(ASTNode):
80     """Built-in print statement.
81
82     Example:  print(x);
83     """
84     expr: ASTNode
85
86 @dataclass
87 class Return(ASTNode):
88     """Return statement inside a function body.
89
90     Example:  return result;
91     """
92     expr: ASTNode
93
94 @dataclass
95 class Block(ASTNode):
96     """A brace-enclosed sequence of statements.
97
98     Example:  { x = 1; print(x); }
99     """
100     statements: list[ASTNode] = field(default_factory=list)

```

```

101
102
103 @dataclass
104 class If(ASTNode):
105     """If / optional else control flow.
106
107     Example:  if (x != 0) { print(x); } else { print(0); }
108     """
109     condition: ASTNode
110     then_block: Block
111     else_block: Optional[Block]
112
113
114 @dataclass
115 class While(ASTNode):
116     """While loop.
117
118     Example:  while (x != 0) { x = x - 1; }
119     """
120     condition: ASTNode
121     body: Block
122
123
124 @dataclass
125 class FunctionDef(ASTNode):
126     """Function definition.
127
128     Example:  def add(a, b) { return a + b; }
129
130     Note: parameter types are unknown at parse time -- Member 3's type
131           checker
132           will infer them from usage. We store only the parameter names here.
133     """
134     name: str
135     params: list[str]
136     body: Block
137
138 #
139 # -----
140 #
141
142 # Top-level
143 #
144 # -----
145
146 @dataclass
147 class Program(ASTNode):
148     """Root node: the entire source file as a list of statements."""
149     statements: list[ASTNode] = field(default_factory=list)

```

Listing 8.5: components/ast_nodes.py — AST node definitions

```

1 from __future__ import annotations
2
3 from typing import Optional
4
5 from components.tokens import Token, TokenType
6 from components.ast_nodes import (
7     ASTNode,
8     Program,
9     Block,
10    Assign,
11    If,
12    While,
13    FunctionDef,
14    FunctionCall,
15    Print,
16    Return,
17    BinaryOp,
18    Literal,
19    Identifier,
20 )
21
22
23 #
24 # -----
25
26 # Parser error
27 #
28 # -----
29
30
31 class ParseError(Exception):
32     pass
33
34 #
35 # -----
36
37 # Parser
38 #
39 # -----
40
41
42 class Parser:
43     """Recursive-descent parser.
44
45     Usage:
46         tokens = Lexer(source).tokenize()
47         tree   = Parser(tokens).parse()    # returns Program node
48     """
49
50     def __init__(self, tokens: list[Token]) -> None:
51         self._tokens = tokens

```

```

45     self._pos = 0
46
47     #
48     -----
49
50     # Public entry point
51     #
52     -----
53
54     def parse(self) -> Program:
55         """Parse the full token stream and return the root Program node
56         ."""
57         line, col = self._peek().line, self._peek().column
58         statements: list[ASTNode] = []
59         while not self._is_at_end():
60             statements.append(self._statement())
61         return Program(statements=statements, line=line, column=col)
62
63     #
64     -----
65
66     # Statements
67     #
68     -----
69
70     def _statement(self) -> ASTNode:
71         """Dispatch to the correct statement parser based on the
72         current token."""
73         tok = self._peek()
74
75         if tok.token_type == TokenType.DEF:
76             return self._func_def()
77
78         if tok.token_type == TokenType.IF:
79             return self._if_stmt()
80
81         if tok.token_type == TokenType.WHILE:
82             return self._while_stmt()
83
84         if tok.token_type == TokenType.RETURN:
85             return self._return_stmt()
86
87         if tok.token_type == TokenType.PRINT:
88             return self._print_stmt()
89
90         # assignment: IDENTIFIER "=" expr ";"
91         if (
92             tok.token_type == TokenType.IDENTIFIER
93             and self._peek_next().token_type == TokenType.ASSIGN
94         ):
95             return self._assignment()

```

```

88
89     # fallback: bare expression statement  expr ";"
90     node = self._expr()
91     self._expect(TokenType.SEMICOLON, "Expected ';' after
92         expression")
93     return node
94
95 def _assignment(self) -> Assign:
96     """IDENTIFIER '=' expr ';' """
97     name_tok = self._advance() #
98     IDENTIFIER
99     self._advance() # '='
100     value = self._expr()
101     self._expect(TokenType.SEMICOLON, "Expected ';' after
102         assignment")
103     return Assign(name=name_tok.lexeme, value=value,
104                   line=name_tok.line, column=name_tok.column)
105
106 def _if_stmt(self) -> If:
107     """'if' '(' expr ')' block ( 'else' block )? """
108     tok = self._advance() # 'if'
109     self._expect(TokenType.LPAREN, "Expected '(' after 'if'")
110     condition = self._expr()
111     self._expect(TokenType.RPAREN, "Expected ')' after if condition
112         ")
113     then_block = self._block()
114     else_block: Optional[Block] = None
115     if self._check(TokenType.ELSE):
116         self._advance() # 'else'
117         else_block = self._block()
118     return If(condition=condition, then_block=then_block,
119              else_block=else_block,
120              line=tok.line, column=tok.column)
121
122 def _while_stmt(self) -> While:
123     """'while' '(' expr ')' block """
124     tok = self._advance() # '
125     while '
126     self._expect(TokenType.LPAREN, "Expected '(' after 'while'")
127     condition = self._expr()
128     self._expect(TokenType.RPAREN, "Expected ')' after while
129         condition")
130     body = self._block()
131     return While(condition=condition, body=body,
132                  line=tok.line, column=tok.column)
133
134 def _func_def(self) -> FunctionDef:
135     """'def' IDENTIFIER '(' params? ')' block """
136     tok = self._advance() # 'def'
137     name_tok = self._expect(TokenType.IDENTIFIER, "Expected
138         function name after 'def'")
139     self._expect(TokenType.LPAREN, "Expected '(' after function

```



```

        name")
132     params = self._params()
133     self._expect(TokenType.RPAREN, "Expected ')' after parameters")
134     body = self._block()
135     return FunctionDef(name=name_tok.lexeme, params=params, body=
        body,
136                           line=tok.line, column=tok.column)
137
138 def _print_stmt(self) -> Print:
139     """'print' '(' expr ')' ';'"""
140     tok = self._advance() # '
        print'
141     self._expect(TokenType.LPAREN, "Expected '(' after 'print'")
142     expr = self._expr()
143     self._expect(TokenType.RPAREN, "Expected ')' after print
        expression")
144     self._expect(TokenType.SEMICOLON, "Expected ';' after print
        statement")
145     return Print(expr=expr, line=tok.line, column=tok.column)
146
147 def _return_stmt(self) -> Return:
148     """'return' expr ';'"""
149     tok = self._advance() # '
        return'
150     expr = self._expr()
151     self._expect(TokenType.SEMICOLON, "Expected ';' after return
        value")
152     return Return(expr=expr, line=tok.line, column=tok.column)
153
154 def _block(self) -> Block:
155     """'{', statement* '}'"""
156     tok = self._expect(TokenType.LBRACE, "Expected '{'")
157     statements: list[ASTNode] = []
158     while not self._check(TokenType.RBRACE) and not self._is_at_end
        ():
159         statements.append(self._statement())
160     self._expect(TokenType.RBRACE, "Expected '}' to close block")
161     return Block(statements=statements, line=tok.line, column=tok.
        column)
162
163 #
        -----
164 # Parameters (function definition)
165 #
        -----
166
167 def _params(self) -> list[str]:
168     """IDENTIFIER (',' IDENTIFIER)* -- returns list of param names.
        """
169     params: list[str] = []
170     if self._check(TokenType.IDENTIFIER):

```

```

171         params.append(self._advance().lexeme)
172         while self._check(TokenType.COMMA):
173             self._advance() # ',', '
174             tok = self._expect(TokenType.IDENTIFIER, "Expected
175                 parameter name after ','")
176             params.append(tok.lexeme)
177         return params
178
179 # -----
180 # Expressions (comparison is lowest precedence, then additive,
181 # then term)
182 # -----
183
184 def _expr(self) -> ASTNode:
185     """additive (('==' | '!=') additive)? -- non-associative
186     comparison"""
187     node = self._additive()
188     if self._check(TokenType.EQUAL_EQUAL) or self._check(TokenType.
189         BANG_EQUAL):
190         op_tok = self._advance()
191         right = self._additive()
192         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
193             line=op_tok.line, column=op_tok.column)
194     return node
195
196 def _additive(self) -> ASTNode:
197     """term (('+' | '-' | '+.' | '-.') term)*"""
198     node = self._term()
199     while self._check(TokenType.PLUS) or self._check(TokenType.
200         MINUS) \
201         or self._check(TokenType.PLUS_DOT) or self._check(
202             TokenType.MINUS_DOT):
203         op_tok = self._advance()
204         right = self._term()
205         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
206             line=op_tok.line, column=op_tok.column)
207     return node
208
209 def _term(self) -> ASTNode:
210     """factor (('*' | '/' | '*.' | '/.') factor)*"""
211     node = self._factor()
212     while self._check(TokenType.STAR) or self._check(TokenType.
213         SLASH) \
214         or self._check(TokenType.STAR_DOT) or self._check(
215             TokenType.SLASH_DOT):
216         op_tok = self._advance()
217         right = self._factor()
218         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
219             line=op_tok.line, column=op_tok.column)

```

```

212     return node
213
214 def _factor(self) -> ASTNode:
215     """
216     '-' factor (unary negation, desugared to 0 - factor)
217     | INT_LITERAL | FLOAT_LITERAL | STRING_LITERAL | BOOL_LITERAL
218     | IDENTIFIER ( '(' args? ')' )?
219     | '(' expr ')'
220     """
221     tok = self._peek()
222
223     # Unary minus '-' factor (integer negation, desugared to 0 -
224     # factor)
225     if tok.token_type == TokenType.MINUS:
226         op_tok = self._advance()
227         operand = self._factor()
228         zero = Literal(value=0, line=op_tok.line, column=op_tok.
229             column)
230         return BinaryOp(op="-", left=zero, right=operand,
231             line=op_tok.line, column=op_tok.column)
232
233     # Unary minus-dot '-.' factor (float negation, desugared to
234     # 0.0 -. factor)
235     if tok.token_type == TokenType.MINUS_DOT:
236         op_tok = self._advance()
237         operand = self._factor()
238         zero_f = Literal(value=0.0, line=op_tok.line, column=op_tok
239             .column)
240         return BinaryOp(op="-.", left=zero_f, right=operand,
241             line=op_tok.line, column=op_tok.column)
242
243     # Integer literal
244     if tok.token_type == TokenType.INT_LITERAL:
245         self._advance()
246         return Literal(value=int(tok.lexeme), line=tok.line, column
247             =tok.column)
248
249     # Float literal
250     if tok.token_type == TokenType.FLOAT_LITERAL:
251         self._advance()
252         return Literal(value=float(tok.lexeme), line=tok.line,
253             column=tok.column)
254
255     # Boolean literal
256     if tok.token_type == TokenType.BOOL_LITERAL:
257         self._advance()
258         return Literal(value=(tok.lexeme == "true"), line=tok.line,
259             column=tok.column)
260
261     # String literal (keep the surrounding quotes stripped)
262     if tok.token_type == TokenType.STRING_LITERAL:
263         self._advance()
264         return Literal(value=tok.lexeme[1:-1], line=tok.line,
265             column=tok.column)

```

```

        column=tok.column)

258
259 # Identifier -- could be a plain variable or a function call
260 if tok.token_type == TokenType.IDENTIFIER:
261     self._advance()
262     if self._check(TokenType.LPAREN):
263         # function call
264         self._advance() # '('
265         args = self._args()
266         self._expect(TokenType.RPAREN, "Expected ')' after
            function arguments")
267         return FunctionCall(name=tok.lexeme, args=args,
268                             line=tok.line, column=tok.column)
269     return Identifier(name=tok.lexeme, line=tok.line, column=
        tok.column)

270
271 # Grouped expression '(' expr ')'
272 if tok.token_type == TokenType.LPAREN:
273     self._advance() # '('
274     node = self._expr()
275     self._expect(TokenType.RPAREN, "Expected ')' to close
        grouped expression")
276     return node
277
278     raise ParseError(self._error_at(tok, f"Unexpected token: {tok.
        lexeme!r}"))
279
280 #
281 -----
282
283 # Arguments (function call)
284 #
285 -----
286
287 def _args(self) -> list[ASTNode]:
288     """expr (',' expr)*"""
289     args: list[ASTNode] = []
290     if not self._check(TokenType.RPAREN):
291         args.append(self._expr())
292         while self._check(TokenType.COMMA):
293             self._advance() # ','
294             args.append(self._expr())
295         return args
296
297 #
298 -----
299
300 # Token navigation helpers
301 #
302 -----

```

```

298 def _peek(self) -> Token:
299     return self._tokens[self._pos]
300
301 def _peek_next(self) -> Token:
302     if self._pos + 1 < len(self._tokens):
303         return self._tokens[self._pos + 1]
304     return self._tokens[-1] # EOF
305
306 def _advance(self) -> Token:
307     tok = self._tokens[self._pos]
308     if not self._is_at_end():
309         self._pos += 1
310     return tok
311
312 def _check(self, token_type: TokenType) -> bool:
313     return self._peek().token_type == token_type
314
315 def _is_at_end(self) -> bool:
316     return self._peek().token_type == TokenType.EOF
317
318 def _expect(self, token_type: TokenType, message: str) -> Token:
319     if self._check(token_type):
320         return self._advance()
321     raise ParseError(self._error_at(self._peek(), message))
322
323 def _error_at(self, tok: Token, message: str) -> str:
324     return f"[line {tok.line}, col {tok.column}] ParseError: {
        message}"

```

Listing 8.6: components/parser.py — recursive-descent parser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from enum import Enum
5
6
7 class LanguageType(Enum):
8     INTEGER = "Integer"
9     FLOAT = "Float"
10    BOOLEAN = "Boolean"
11    STRING = "String"
12    VOID = "Void"
13
14
15 @dataclass
16 class Symbol:
17     name: str
18     symbol_type: LanguageType
19
20
21 @dataclass
22 class VariableSymbol(Symbol):

```

```

23     initialized: bool = False
24
25
26 @dataclass
27 class FunctionSymbol(Symbol):
28     parameters: list[tuple[str, LanguageType]] = field(default_factory=
        list)
29
30
31 class SymbolTableError(Exception):
32     pass
33
34
35 class SymbolTable:
36     def __init__(self, parent: SymbolTable | None = None) -> None:
37         self.parent = parent
38         self._symbols: dict[str, Symbol] = {}
39
40     def define_variable(self, name: str, symbol_type: LanguageType,
        initialized: bool = False) -> VariableSymbol:
41         if name in self._symbols:
42             raise SymbolTableError(f"Symbol already defined in this
                scope: {name}")
43         symbol = VariableSymbol(name=name, symbol_type=symbol_type,
            initialized=initialized)
44         self._symbols[name] = symbol
45         return symbol
46
47     def define_function(
48         self,
49         name: str,
50         return_type: LanguageType,
51         parameters: list[tuple[str, LanguageType]],
52     ) -> FunctionSymbol:
53         if name in self._symbols:
54             raise SymbolTableError(f"Symbol already defined in this
                scope: {name}")
55         symbol = FunctionSymbol(name=name, symbol_type=return_type,
            parameters=parameters)
56         self._symbols[name] = symbol
57         return symbol
58
59     def lookup(self, name: str) -> Symbol:
60         symbol = self._symbols.get(name)
61         if symbol is not None:
62             return symbol
63         if self.parent is not None:
64             return self.parent.lookup(name)
65         raise SymbolTableError(f"Undefined symbol: {name}")
66
67     def exists_in_current_scope(self, name: str) -> bool:
68         return name in self._symbols
69

```

```

70 def child_scope(self) -> SymbolTable:
71     return SymbolTable(parent=self)
72
73 def set_initialized(self, name: str) -> None:
74     symbol = self.lookup(name)
75     if not isinstance(symbol, VariableSymbol):
76         raise SymbolTableError(f"Cannot initialize non-variable
77                                 symbol: {name}")
78     symbol.initialized = True
79
80 def snapshot(self) -> dict[str, Symbol]:
81     return dict(self._symbols)
82
83 def format_table(self) -> str:
84     lines = [
85         f"{'Name':<12} {'Kind':<10} {'Type':<10} {'Initialized'
86         ':<12} Parameters",
87         "-" * 72,
88     ]
89
90     for name, symbol in self._symbols.items():
91         if isinstance(symbol, FunctionSymbol):
92             params = ", ".join(f"{param_name}: {param_type.value}"
93                                 for param_name, param_type in symbol.parameters)
94             lines.append(
95                 f"{name:<12} {'function':<10} {symbol.symbol_type.
96                 value:<10} {'-':<12} {params or '-'}"
97             )
98         elif isinstance(symbol, VariableSymbol):
99             initialized = "yes" if symbol.initialized else "no"
100             lines.append(
101                 f"{name:<12} {'variable':<10} {symbol.symbol_type.
102                 value:<10} {initialized:<12} -"
103             )
104         else:
105             lines.append(f"{name:<12} {'symbol':<10} {symbol.
106                             symbol_type.value:<10} {'-':<12} -")
107
108     return "\n".join(lines)

```

Listing 8.7: components/symbol_table.py — scoped symbol table

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4
5 from components.ast_nodes import (
6     ASTNode,
7     Assign,
8     BinaryOp,
9     Block,
10    FunctionCall,
11    FunctionDef,

```

```

12     Identifier,
13     If,
14     Literal,
15     Print,
16     Program,
17     Return,
18     While,
19 )
20 from components.symbol_table import FunctionSymbol, LanguageType,
    SymbolTable, SymbolTableError, VariableSymbol
21
22
23 class TypeCheckError(Exception):
24     pass
25
26
27 @dataclass
28 class _FunctionState:
29     node: FunctionDef
30     body_checked: bool = False
31
32
33 @dataclass
34 class _FunctionContext:
35     name: str
36     return_type: LanguageType | None = None
37
38
39 class TypeChecker:
40     def __init__(self) -> None:
41         self.global_scope = SymbolTable()
42         self._functions: dict[str, _FunctionState] = {}
43         self._active_calls: list[str] = []
44
45     def check(self, program: Program) -> SymbolTable:
46         for statement in program.statements:
47             if isinstance(statement, FunctionDef):
48                 self._register_function(statement)
49
50         for statement in program.statements:
51             if isinstance(statement, FunctionDef):
52                 continue
53             self._check_statement(statement, self.global_scope, None)
54
55         self._check_uncalled_functions()
56         return self.global_scope
57
58     def _register_function(self, node: FunctionDef) -> None:
59         if node.name in self._functions:
60             raise TypeCheckError(self._error_at(node, f"Function '{node
                .name}' is already defined"))
61
62         placeholder_parameters = [(param_name, LanguageType.VOID) for

```



```

        param_name in node.params]
63     try:
64         self.global_scope.define_function(node.name, LanguageType.
            VOID, placeholder_parameters)
65     except SymbolTableError as error:
66         raise TypeCheckError(self._error_at(node, str(error))) from
            error
67     self._functions[node.name] = _FunctionState(node=node)
68
69     def _check_statement(
70         self,
71         node: ASTNode,
72         scope: SymbolTable,
73         function_context: _FunctionContext | None,
74     ) -> None:
75         if isinstance(node, Assign):
76             value_type = self._infer_expr_type(node.value, scope)
77             self._assign_variable_type(scope, node, value_type)
78             return
79
80         if isinstance(node, Print):
81             self._infer_expr_type(node.expr, scope)
82             return
83
84         if isinstance(node, Return):
85             if function_context is None:
86                 raise TypeCheckError(self._error_at(node, "'return' is
                    only valid inside a function"))
87             expr_type = self._infer_expr_type(node.expr, scope)
88             if function_context.return_type is None:
89                 function_context.return_type = expr_type
90             elif function_context.return_type != expr_type:
91                 raise TypeCheckError(
92                     self._error_at(
93                         node,
94                         f"Function '{function_context.name}' returns
                            both {function_context.return_type.value}
                            and {expr_type.value}",
95                     )
96                 )
97             return
98
99         if isinstance(node, If):
100             condition_type = self._infer_expr_type(node.condition,
                scope)
101             if condition_type != LanguageType.BOOLEAN:
102                 raise TypeCheckError(self._error_at(node.condition, "If
                    condition must have type Boolean"))
103             self._check_block(node.then_block, scope.child_scope(),
                function_context)
104             if node.else_block is not None:
105                 self._check_block(node.else_block, scope.child_scope(),
                    function_context)

```

```

106         return
107
108     if isinstance(node, While):
109         condition_type = self._infer_expr_type(node.condition,
110                                                 scope)
111         if condition_type != LanguageType.BOOLEAN:
112             raise TypeCheckError(self._error_at(node.condition, "
113                 While condition must have type Boolean"))
114         self._check_block(node.body, scope.child_scope(),
115                           function_context)
116         return
117
118     if isinstance(node, Block):
119         self._check_block(node, scope.child_scope(),
120                           function_context)
121         return
122
123     if isinstance(node, FunctionCall):
124         self._infer_function_call_type(node, scope)
125         return
126
127     raise TypeCheckError(self._error_at(node, f"Unsupported
128         statement node: {type(node).__name__}"))
129
130 def _check_block(
131     self,
132     block: Block,
133     scope: SymbolTable,
134     function_context: _FunctionContext | None,
135 ) -> None:
136     for statement in block.statements:
137         self._check_statement(statement, scope, function_context)
138
139 def _infer_expr_type(self, node: ASTNode, scope: SymbolTable) ->
140 LanguageType:
141     if isinstance(node, Literal):
142         return self._literal_type(node)
143
144     if isinstance(node, Identifier):
145         try:
146             symbol = scope.lookup(node.name)
147         except SymbolTableError as error:
148             raise TypeCheckError(self._error_at(node, str(error)))
149             from error
150         if not isinstance(symbol, VariableSymbol):
151             raise TypeCheckError(self._error_at(node, f"'{node.name
152                 }' is not a variable"))
153         return symbol.symbol_type
154
155     if isinstance(node, FunctionCall):
156         return self._infer_function_call_type(node, scope)
157
158     if isinstance(node, BinaryOp):

```

```

151         left_type = self._infer_expr_type(node.left, scope)
152         right_type = self._infer_expr_type(node.right, scope)
153         return self._infer_binary_type(node, left_type, right_type)
154
155         raise TypeCheckError(self._error_at(node, f"Unsupported
156         expression node: {type(node).__name__}"))
157
158     def _infer_function_call_type(self, node: FunctionCall, scope:
159     SymbolTable) -> LanguageType:
160         function_state = self._functions.get(node.name)
161         if function_state is None:
162             raise TypeCheckError(self._error_at(node, f"Undefined
163             function: {node.name}"))
164         if node.name in self._active_calls:
165             raise TypeCheckError(self._error_at(node, f"Recursive
166             function calls are not supported: {node.name}"))
167
168         argument_types = [self._infer_expr_type(argument, scope) for
169         argument in node.args]
170         function_symbol = self._lookup_function_symbol(node)
171
172         if len(argument_types) != len(function_symbol.parameters):
173             raise TypeCheckError(
174                 self._error_at(
175                     node,
176                     f"Function '{node.name}' expects {len(
177                         function_symbol.parameters)} argument(s), got {
178                         len(argument_types)}",
179                 )
180             )
181
182         declared_parameter_types = [param_type for _, param_type in
183         function_symbol.parameters]
184         if all(param_type == LanguageType.VOID for param_type in
185         declared_parameter_types):
186             function_symbol.parameters = list(zip(function_state.node.
187             params, argument_types, strict=False))
188         else:
189             for index, ((_, expected_type), actual_type) in enumerate(
190                 zip(function_symbol.parameters, argument_types, strict=
191                 False)):
192                 if expected_type != actual_type:
193                     raise TypeCheckError(
194                         self._error_at(
195                             node.args[index],
196                             f"Argument {index + 1} of '{node.name}'
197                             must be {expected_type.value}, got {
198                                 actual_type.value}",
199                         )
200                     )
201
202         if not function_state.body_checked:
203             self._check_function_body(function_state, function_symbol)

```

```

190     return function_symbol.symbol_type
191
192
193 def _check_function_body(self, function_state: _FunctionState,
194 function_symbol: FunctionSymbol) -> None:
195     function_context = _FunctionContext(name=function_state.node.
196 name)
197     function_scope = self.global_scope.child_scope()
198     for parameter_name, parameter_type in function_symbol.
199 parameters:
200         function_scope.define_variable(parameter_name,
201 parameter_type, initialized=True)
202
203 self._active_calls.append(function_state.node.name)
204 try:
205     self._check_block(function_state.node.body, function_scope,
206 function_context)
207 finally:
208     self._active_calls.pop()
209
210 function_symbol.symbol_type = function_context.return_type or
211 LanguageType.VOID
212 function_state.body_checked = True
213
214 def _check_uncalled_functions(self) -> None:
215     """After all call-site checks, type-check any function whose
216 body was never visited."""
217     for name, state in self._functions.items():
218         if not state.body_checked:
219             try:
220                 function_symbol = self.global_scope.lookup(name)
221             except SymbolTableError:
222                 continue
223             if not isinstance(function_symbol, FunctionSymbol):
224                 continue
225             inferred_params = self._infer_param_types_from_body(
226 state.node)
227             function_symbol.parameters = inferred_params
228             self._check_function_body(state, function_symbol)
229
230 def _infer_param_types_from_body(self, node: FunctionDef) -> list[
231 tuple[str, LanguageType]]:
232     """Infer parameter types by scanning the body for expression
233 contexts.
234
235     Any parameter used as an operand of an arithmetic or comparison
236 operator
237 is inferred as numeric (Float if the sibling operand is a float
238 literal,
239 Integer otherwise). Parameters whose usage gives no type hint
240 default to
241 Integer.
242 """

```

```

230 param_names = set(node.params)
231 inferred: dict[str, LanguageType] = {}
232
233 def scan_expr(n: ASTNode) -> None:
234     if isinstance(n, BinaryOp):
235         if n.op in {"+", "-", "*", "/", "=", "!="}:
236             for operand in (n.left, n.right):
237                 if isinstance(operand, Identifier) and operand.
238                     name in param_names:
239                         inferred.setdefault(operand.name,
240                             LanguageType.INTEGER)
241             elif n.op in {"+.", "-.", "*.", "/."}:
242                 for operand in (n.left, n.right):
243                     if isinstance(operand, Identifier) and operand.
244                         name in param_names:
245                             inferred.setdefault(operand.name,
246                                 LanguageType.FLOAT)
247             scan_expr(n.left)
248             scan_expr(n.right)
249         elif isinstance(n, FunctionCall):
250             for arg in n.args:
251                 scan_expr(arg)
252
253 def scan_stmt(n: ASTNode) -> None:
254     if isinstance(n, Assign):
255         scan_expr(n.value)
256     elif isinstance(n, Print):
257         scan_expr(n.expr)
258     elif isinstance(n, Return):
259         scan_expr(n.expr)
260     elif isinstance(n, If):
261         scan_expr(n.condition)
262         scan_block(n.then_block)
263         if n.else_block:
264             scan_block(n.else_block)
265     elif isinstance(n, While):
266         scan_expr(n.condition)
267         scan_block(n.body)
268     elif isinstance(n, Block):
269         scan_block(n)
270     elif isinstance(n, FunctionCall):
271         for arg in n.args:
272             scan_expr(arg)
273
274 def scan_block(block: Block) -> None:
275     for stmt in block.statements:
276         scan_stmt(stmt)
277
278 scan_block(node.body)
279 return [(param, inferred.get(param, LanguageType.INTEGER)) for
280         param in node.params]
281
282 def _assign_variable_type(self, scope: SymbolTable, node: Assign,

```

```

value_type: LanguageType) -> None:
278     try:
279         symbol = scope.lookup(node.name)
280     except SymbolTableError:
281         scope.define_variable(node.name, value_type, initialized=
            True)
282         return
283
284     if not isinstance(symbol, VariableSymbol):
285         raise TypeCheckError(self._error_at(node, f"'{node.name}',
            is not a variable"))
286
287     if symbol.symbol_type != value_type:
288         raise TypeCheckError(
289             self._error_at(
290                 node,
291                 f"Cannot assign {value_type.value} to variable '{
                    node.name}' of type {symbol.symbol_type.value}",
292             )
293         )
294     symbol.initialized = True
295
296 def _infer_binary_type(
297     self,
298     node: BinaryOp,
299     left_type: LanguageType,
300     right_type: LanguageType,
301 ) -> LanguageType:
302     # Integer operators: both operands must be Integer
303     if node.op in {"+", "-", "*"}:
304         if left_type != LanguageType.INTEGER or right_type !=
            LanguageType.INTEGER:
305             raise TypeCheckError(
306                 self._error_at(node, f"Operator '{node.op}'
                    requires two Integer operands")
307             )
308         return LanguageType.INTEGER
309
310     # Integer division: both operands must be Integer, result is
        Integer
311     if node.op == "/":
312         if left_type != LanguageType.INTEGER or right_type !=
            LanguageType.INTEGER:
313             raise TypeCheckError(
314                 self._error_at(node, f"Operator '{node.op}'
                    requires two Integer operands")
315             )
316         return LanguageType.INTEGER
317
318     # Float operators: both operands must be Float
319     if node.op in {"+.", "-.", "*.", "/."}:
320         if left_type != LanguageType.FLOAT or right_type !=
            LanguageType.FLOAT:
            raise TypeCheckError(

```

```

321         self._error_at(node, f"Operator '{node.op}'
322                             requires two Float operands")
323     )
324     return LanguageType.FLOAT
325
326     if node.op in {"==", "!="}:
327         if not self._is_numeric(left_type) or not self._is_numeric(
328             right_type):
329             raise TypeCheckError(
330                 self._error_at(
331                     node,
332                     f"Operator '{node.op}' requires two arithmetic
333                     operands",
334                 )
335             )
336         if left_type != right_type:
337             raise TypeCheckError(
338                 self._error_at(
339                     node,
340                     f"Operator '{node.op}' requires operands of the
341                     same type, got {left_type.value} and {
342                         right_type.value}",
343                 )
344             )
345         return LanguageType.BOOLEAN
346
347     raise TypeCheckError(self._error_at(node, f"Unsupported
348         operator: {node.op}"))
349
350 def _lookup_function_symbol(self, node: FunctionCall) ->
351     FunctionSymbol:
352     try:
353         symbol = self.global_scope.lookup(node.name)
354     except SymbolTableError as error:
355         raise TypeCheckError(self._error_at(node, str(error))) from
356             error
357     if not isinstance(symbol, FunctionSymbol):
358         raise TypeCheckError(self._error_at(node, f"'{node.name}'
359             is not a function"))
360     return symbol
361
362 @staticmethod
363 def _is_numeric(language_type: LanguageType) -> bool:
364     return language_type in {LanguageType.INTEGER, LanguageType.
365         FLOAT}
366
367 @staticmethod
368 def _literal_type(node: Literal) -> LanguageType:
369     if isinstance(node.value, bool):
370         return LanguageType.BOOLEAN
371     if isinstance(node.value, int):
372         return LanguageType.INTEGER
373     if isinstance(node.value, float):

```

```

364         return LanguageType.FLOAT
365     if isinstance(node.value, str):
366         return LanguageType.STRING
367     raise TypeCheckError(f"Unsupported literal value: {node.value!r}
368                             ")
369
370     @staticmethod
371     def _error_at(node: ASTNode, message: str) -> str:
372         return f"[line {node.line}, col {node.column}] TypeError: {
373             message}"

```

Listing 8.8: components/type_checker.py — static type checker with inference

```

1  from __future__ import annotations
2
3  from dataclasses import dataclass
4  from typing import Callable
5
6  from components.ast_nodes import (
7      ASTNode,
8      Assign,
9      BinaryOp,
10     Block,
11     FunctionCall,
12     FunctionDef,
13     Identifier,
14     If,
15     Literal,
16     Print,
17     Program,
18     Return,
19     While,
20 )
21 from components.symbol_table import LanguageType
22
23
24 class InterpreterError(Exception):
25     pass
26
27
28 class _ReturnSignal(Exception):
29     def __init__(self, value: object) -> None:
30         self.value = value
31
32
33 @dataclass
34 class _FunctionRuntime:
35     node: FunctionDef
36
37
38 class _Environment:
39     def __init__(self, parent: _Environment | None = None) -> None:
40         self.parent = parent

```



```

41     self.values: dict[str, object] = {}
42
43     def define(self, name: str, value: object) -> None:
44         self.values[name] = value
45
46     def get(self, name: str) -> object:
47         if name in self.values:
48             return self.values[name]
49         if self.parent is not None:
50             return self.parent.get(name)
51         raise InterpreterError(f"Undefined variable: {name}")
52
53     def assign(self, name: str, value: object) -> None:
54         if name in self.values:
55             self.values[name] = value
56             return
57         if self.parent is not None and self.parent.contains(name):
58             self.parent.assign(name, value)
59             return
60         self.values[name] = value
61
62     def contains(self, name: str) -> bool:
63         if name in self.values:
64             return True
65         if self.parent is not None:
66             return self.parent.contains(name)
67         return False
68
69
70 class Interpreter:
71     def __init__(self, output_callback: Callable[[str], None] | None =
None) -> None:
72         self._functions: dict[str, _FunctionRuntime] = {}
73         self._output_callback = output_callback or print
74
75     def execute(self, program: Program) -> None:
76         environment = _Environment()
77         for statement in program.statements:
78             if isinstance(statement, FunctionDef):
79                 self._functions[statement.name] = _FunctionRuntime(node
=statement)
80
81         for statement in program.statements:
82             if isinstance(statement, FunctionDef):
83                 continue
84             self._execute_statement(statement, environment)
85
86     def _execute_statement(self, node: ASTNode, environment:
_Environment) -> None:
87         if isinstance(node, Assign):
88             value = self._evaluate(node.value, environment)
89             environment.assign(node.name, value)
90             return

```

```

91
92     if isinstance(node, Print):
93         value = self._evaluate(node.expr, environment)
94         self._output_callback(f"{self._format_value(value)} : {self
95             ._runtime_type(value).value}")
96         return
97
98     if isinstance(node, Return):
99         raise _ReturnSignal(self._evaluate(node.expr, environment))
100
101     if isinstance(node, If):
102         condition = self._evaluate(node.condition, environment)
103         if not isinstance(condition, bool):
104             raise InterpreterError(self._error_at(node.condition, "
105                 If condition must evaluate to Boolean"))
106         branch = node.then_block if condition else node.else_block
107         if branch is not None:
108             self._execute_block(branch, _Environment(parent=
109                 environment))
110         return
111
112     if isinstance(node, While):
113         while True:
114             condition = self._evaluate(node.condition, environment)
115             if not isinstance(condition, bool):
116                 raise InterpreterError(self._error_at(node.
117                     condition, "While condition must evaluate to
118                     Boolean"))
119             if not condition:
120                 break
121             self._execute_block(node.body, _Environment(parent=
122                 environment))
123         return
124
125     if isinstance(node, Block):
126         self._execute_block(node, _Environment(parent=environment))
127         return
128
129     if isinstance(node, FunctionCall):
130         self._evaluate(node, environment)
131         return
132
133     raise InterpreterError(self._error_at(node, f"Unsupported
134         statement node: {type(node).__name__}"))
135
136 def _execute_block(self, block: Block, environment: _Environment)
137     -> None:
138     for statement in block.statements:
139         self._execute_statement(statement, environment)
140
141 def _evaluate(self, node: ASTNode, environment: _Environment) ->
142     object:
143     if isinstance(node, Literal):

```

```

135         return node.value
136
137     if isinstance(node, Identifier):
138         return environment.get(node.name)
139
140     if isinstance(node, BinaryOp):
141         left_value = self._evaluate(node.left, environment)
142         right_value = self._evaluate(node.right, environment)
143         return self._evaluate_binary(node, left_value, right_value)
144
145     if isinstance(node, FunctionCall):
146         return self._call_function(node, environment)
147
148     raise InterpreterError(self._error_at(node, f"Unsupported
149         expression node: {type(node).__name__}"))
150
151 def _call_function(self, node: FunctionCall, environment:
152     _Environment) -> object:
153     function_runtime = self._functions.get(node.name)
154     if function_runtime is None:
155         raise InterpreterError(self._error_at(node, f"Undefined
156             function: {node.name}"))
157
158     if len(node.args) != len(function_runtime.node.params):
159         raise InterpreterError(
160             self._error_at(
161                 node,
162                 f"Function '{node.name}' expects {len(
163                     function_runtime.node.params)} argument(s), got
164                     {len(node.args)}",
165             )
166         )
167
168     call_environment = _Environment(parent=environment)
169     argument_values = [self._evaluate(argument, environment) for
170         argument in node.args]
171     for parameter_name, argument_value in zip(function_runtime.node
172         .params, argument_values, strict=False):
173         call_environment.define(parameter_name, argument_value)
174
175     try:
176         self._execute_block(function_runtime.node.body,
177             call_environment)
178     except _ReturnSignal as signal:
179         return signal.value
180     return None
181
182 def _evaluate_binary(self, node: BinaryOp, left_value: object,
183     right_value: object) -> object:
184     if node.op == "+":
185         return left_value + right_value
186     if node.op == "-":
187         return left_value - right_value

```

```

179     if node.op == "*":
180         return left_value * right_value
181     if node.op == "/":
182         if int(right_value) == 0:
183             raise InterpreterError(self._error_at(node, "Division
184                                     by zero"))
185         return int(left_value) // int(right_value)
186     if node.op == "+":
187         return float(left_value) + float(right_value)
188     if node.op == "-":
189         return float(left_value) - float(right_value)
190     if node.op == "*":
191         return float(left_value) * float(right_value)
192     if node.op == "/":
193         if float(right_value) == 0.0:
194             raise InterpreterError(self._error_at(node, "Division
195                                     by zero"))
196         return float(left_value) / float(right_value)
197     if node.op == "==":
198         return left_value == right_value
199     if node.op == "!=":
200         return left_value != right_value
201     raise InterpreterError(self._error_at(node, f"Unsupported
202                                     operator: {node.op}"))
203
204 @staticmethod
205 def _format_value(value: object) -> str:
206     if isinstance(value, str):
207         return f'"{value}"'
208     if isinstance(value, bool):
209         return "true" if value else "false"
210     return str(value)
211
212 @staticmethod
213 def _runtime_type(value: object) -> LanguageType:
214     if isinstance(value, bool):
215         return LanguageType.BOOLEAN
216     if isinstance(value, int):
217         return LanguageType.INTEGER
218     if isinstance(value, float):
219         return LanguageType.FLOAT
220     if isinstance(value, str):
221         return LanguageType.STRING
222     return LanguageType.VOID
223
224 @staticmethod
225 def _error_at(node: ASTNode, message: str) -> str:
226     return f"[line {node.line}, col {node.column}] RuntimeError: {
227         message}"

```

Listing 8.9: components/interpreter.py — tree-walking interpreter

```

1 from __future__ import annotations

```

```

2
3 from dataclasses import dataclass
4
5 from components.ast_printer import ASTPrinter
6 from components.interpreter import Interpreter
7 from components.lexica import Lexer
8 from components.parser import Parser
9 from components.symbol_table import SymbolTable
10 from components.tokens import Token
11 from components.type_checker import TypeChecker
12
13
14 @dataclass
15 class PipelineResult:
16     tokens: list[Token]
17     ast_text: str
18     checked_scope: SymbolTable
19     outputs: list[str]
20
21
22 def run_pipeline(source_code: str) -> PipelineResult:
23     tokens = Lexer(source_code).tokenize()
24     tree = Parser(tokens).parse()
25     ast_text = ASTPrinter().print(tree)
26     checked_scope = TypeChecker().check(tree)
27
28     outputs: list[str] = []
29     Interpreter(output_callback=outputs.append).execute(tree)
30
31     return PipelineResult(
32         tokens=tokens,
33         ast_text=ast_text,
34         checked_scope=checked_scope,
35         outputs=outputs,
36     )
37
38
39 def format_tokens(tokens: list[Token]) -> str:
40     lines = []
41     for token in tokens:
42         lines.append(
43             f"    {token.token_type.name:<14} {token.lexeme!r:<12} line={
44                 token.line} col={token.column}"
45         )
46     return "\n".join(lines)
47
48
49 def format_stage_output(result: PipelineResult) -> str:
50     sections = [
51         "=" * 60,
52         "  STAGE 1 -- TOKENS",
53         "=" * 60,
54         format_tokens(result.tokens),
55     ]

```

```

54     "",
55     "=" * 60,
56     "    STAGE 2 -- AST (parser output)",
57     "=" * 60,
58     result.ast_text,
59     "",
60     "=" * 60,
61     "    STAGE 3 -- TYPE CHECK",
62     "=" * 60,
63     result.checked_scope.format_table(),
64     "",
65     "=" * 60,
66     "    STAGE 4 -- EXECUTION",
67     "=" * 60,
68     "\n".join(result.outputs) if result.outputs else "(no output)",
69 ]
70 return "\n".join(sections)

```

Listing 8.10: components/pipeline.py — shared pipeline (CLI, UI, and tests)

A.3 Test Suite

```

1 from __future__ import annotations
2
3 import unittest
4
5 from components.lexica import Lexer, LexerError
6 from components.parser import Parser, ParseError
7 from components.pipeline import run_pipeline
8 from components.symbol_table import LanguageType, SymbolTable,
9     SymbolTableError
10 from components.tokens import TokenType
11 from components.type_checker import TypeCheckError
12
13 #
14 # -----
15 #
16
17 # Lexer
18 # -----
19
20 class LexerTests(unittest.TestCase):
21     """Tests for components/lexica.py -- lexical analysis and
22     tokenisation."""
23
24     def test_integer_literal_token(self) -> None:
25         tokens = Lexer("42").tokenize()
26         self.assertEqual(tokens[0].token_type, TokenType.INT_LITERAL)
27         self.assertEqual(tokens[0].lexeme, "42")

```

```

25 def test_float_literal_token(self) -> None:
26     tokens = Lexer("3.14").tokenize()
27     self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
28     self.assertEqual(tokens[0].lexeme, "3.14")
29
30 def test_bool_literals_produce_bool_literal_tokens(self) -> None:
31     tokens = Lexer("true false").tokenize()
32     self.assertEqual(tokens[0].token_type, TokenType.BOOL_LITERAL)
33     self.assertEqual(tokens[0].lexeme, "true")
34     self.assertEqual(tokens[1].token_type, TokenType.BOOL_LITERAL)
35     self.assertEqual(tokens[1].lexeme, "false")
36
37 def test_keywords_produce_correct_token_types(self) -> None:
38     tokens = Lexer("if else while def return print").tokenize()
39     expected = [
40         TokenType.IF, TokenType.ELSE, TokenType.WHILE,
41         TokenType.DEF, TokenType.RETURN, TokenType.PRINT,
42     ]
43     actual = [t.token_type for t in tokens if t.token_type !=
44               TokenType.EOF]
45     self.assertEqual(actual, expected)
46
47 def test_identifier_not_confused_with_keyword(self) -> None:
48     tokens = Lexer("iffy whileloop").tokenize()
49     self.assertEqual(tokens[0].token_type, TokenType.IDENTIFIER)
50     self.assertEqual(tokens[1].token_type, TokenType.IDENTIFIER)
51
52 def test_two_character_operators(self) -> None:
53     tokens = Lexer("== !=").tokenize()
54     self.assertEqual(tokens[0].token_type, TokenType.EQUAL_EQUAL)
55     self.assertEqual(tokens[1].token_type, TokenType.BANG_EQUAL)
56
57 def test_float_dot_operators_tokenised(self) -> None:
58     tokens = Lexer("+. -. *. /.").tokenize()
59     expected = [TokenType.PLUS_DOT, TokenType.MINUS_DOT, TokenType.
60                 STAR_DOT, TokenType.SLASH_DOT]
61     actual = [t.token_type for t in tokens if t.token_type !=
62               TokenType.EOF]
63     self.assertEqual(actual, expected)
64
65 def test_float_literal_not_confused_with_dot_operator(self) -> None
66 :
67     # "3.14" must produce FLOAT_LITERAL, not INT_LITERAL followed
68     # by something
69     tokens = Lexer("3.14").tokenize()
70     self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
71     self.assertEqual(tokens[0].lexeme, "3.14")
72
73 def test_source_position_tracked_across_lines(self) -> None:
74     tokens = Lexer("x\ny").tokenize()
75     self.assertEqual(tokens[0].line, 1)
76     self.assertEqual(tokens[1].line, 2)

```

```

73     def test_unexpected_character_raises_lexer_error(self) -> None:
74         with self.assertRaises(LexerError):
75             Lexer("@").tokenize()
76
77     def test_lone_exclamation_raises_lexer_error(self) -> None:
78         with self.assertRaises(LexerError):
79             Lexer("!5").tokenize()
80
81     def test_terminated_string_raises_lexer_error(self) -> None:
82         with self.assertRaises(LexerError):
83             Lexer('"hello').tokenize()
84
85
86 #
87 # -----
88 # Symbol table
89 # -----
90
91 class SymbolTableTests(unittest.TestCase):
92     """Tests for components/symbol_table.py -- scoped symbol table and
93     semantic types."""
94
95     def test_define_variable_and_lookup(self) -> None:
96         table = SymbolTable()
97         table.define_variable("x", LanguageType.INTEGER, initialized=
98             True)
99         symbol = table.lookup("x")
100         self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
101
102     def test_duplicate_variable_definition_raises_error(self) -> None:
103         table = SymbolTable()
104         table.define_variable("x", LanguageType.INTEGER)
105         with self.assertRaises(SymbolTableError):
106             table.define_variable("x", LanguageType.FLOAT)
107
108     def test_duplicate_function_definition_raises_error(self) -> None:
109         table = SymbolTable()
110         table.define_function("f", LanguageType.INTEGER, [])
111         with self.assertRaises(SymbolTableError):
112             table.define_function("f", LanguageType.FLOAT, [])
113
114     def test_lookup_in_parent_scope(self) -> None:
115         parent = SymbolTable()
116         parent.define_variable("x", LanguageType.INTEGER, initialized=
117             True)
118         child = parent.child_scope()
119         symbol = child.lookup("x")
120         self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
121
122     def test_undefined_symbol_raises_error(self) -> None:

```



```

119     table = SymbolTable()
120     with self.assertRaises(SymbolTableError):
121         table.lookup("z")
122
123     def test_format_table_contains_variable_row(self) -> None:
124         table = SymbolTable()
125         table.define_variable("counter", LanguageType.INTEGER,
126                               initialized=True)
127         output = table.format_table()
128         self.assertIn("counter", output)
129         self.assertIn("variable", output)
130         self.assertIn("Integer", output)
131
132     def test_format_table_contains_function_row(self) -> None:
133         table = SymbolTable()
134         table.define_function("add", LanguageType.INTEGER, [("a",
135                                                             LanguageType.INTEGER)])
136         output = table.format_table()
137         self.assertIn("add", output)
138         self.assertIn("function", output)
139
140     #
141     # Parser
142     #
143
144     class ParserTests(unittest.TestCase):
145         """Tests for components/parser.py -- recursive-descent parsing."""
146
147         def _parse(self, source: str):
148             return Parser(Lexer(source).tokenize()).parse()
149
150         def test_missing_semicolon_raises_parse_error(self) -> None:
151             with self.assertRaises(ParseError):
152                 self._parse("x = 5")
153
154         def test_unclosed_brace_raises_parse_error(self) -> None:
155             with self.assertRaises(ParseError):
156                 self._parse("if (true) { x = 1;")
157
158         def test_operator_precedence_mul_before_add(self) -> None:
159             # 2 + 3 * 4 must be 14, not 20
160             result = run_pipeline("x = 2 + 3 * 4; print(x);")
161             self.assertEqual(result.outputs, ["14 : Integer"])
162
163         def test_operator_left_associativity(self) -> None:
164             # 10 - 3 - 2 must be 5 (left-associative), not 9
165             result = run_pipeline("x = 10 - 3 - 2; print(x);")
166             self.assertEqual(result.outputs, ["5 : Integer"])

```

```

166
167     def test_parentheses_override_precedence(self) -> None:
168         # (2 + 3) * 4 must be 20
169         result = run_pipeline("x = (2 + 3) * 4; print(x);")
170         self.assertEqual(result.outputs, ["20 : Integer"])
171
172
173     #
174     -----
175     # Type checker
176     #
177     -----
178
179     class TypeCheckerTests(unittest.TestCase):
180         """Tests for components/type_checker.py -- static type inference and
181         checking."""
182
183     def test_integer_arithmetic_stays_integer(self) -> None:
184         result = run_pipeline("x = 3 + 4; print(x);")
185         self.assertEqual(result.outputs, ["7 : Integer"])
186
187     def test_division_always_produces_integer(self) -> None:
188         # integer / integer -> Integer (no implicit float conversion)
189         result = run_pipeline("x = 10 / 2; print(x);")
190         self.assertEqual(result.outputs, ["5 : Integer"])
191
192     def test_float_dot_operators_produce_float(self) -> None:
193         result = run_pipeline("x = 1.0 +. 2.5; print(x);")
194         self.assertEqual(result.outputs, ["3.5 : Float"])
195
196     def test_mixed_int_float_addition_raises_type_error(self) -> None:
197         # Integer + Float is no longer valid; must use +.
198         with self.assertRaises(TypeError):
199             run_pipeline("x = 1 + 2.5; print(x);")
200
201     def test_string_variable_type_inferred(self) -> None:
202         result = run_pipeline('x = "hello"; print(x);')
203         self.assertEqual(result.outputs, ['"hello" : String'])
204
205     def test_boolean_literal_type_inferred(self) -> None:
206         result = run_pipeline("x = true; print(x);")
207         self.assertEqual(result.outputs, ["true : Boolean"])
208
209     def test_boolean_equality_raises_type_error(self) -> None:
210         # Boolean == Boolean is not allowed; == only accepts arithmetic
211         # operands
212         with self.assertRaises(TypeError):
213             run_pipeline('x = true; if (x == false) { print("then"); }
214                             else { print("else"); }')
215
216     def test_mixed_int_float_equality_raises_type_error(self) -> None:

```

```

212     # == and != must not accept mixed Integer/Float (no overloading
213     )
214     with self.assertRaises(TypeError):
215         run_pipeline("x = 5; y = 3.14; if (x == y) { print(x); }")
216
217 def test_mixed_int_float_inequality_raises_type_error(self) -> None:
218     :
219     with self.assertRaises(TypeError):
220         run_pipeline("x = 5; y = 3.14; if (x != y) { print(x); }")
221
222 def test_arithmetic_on_string_raises_type_error(self) -> None:
223     with self.assertRaises(TypeError):
224         run_pipeline('x = "a" + 1;')
225
226 def test_if_non_boolean_condition_raises_type_error(self) -> None:
227     with self.assertRaises(TypeError):
228         run_pipeline("if (1) { x = 2; }")
229
230 def test_while_non_boolean_condition_raises_type_error(self) ->
231 None:
232     with self.assertRaises(TypeError):
233         run_pipeline("x = 0; while (x) { x = x + 1; }")
234
235 def test_assignment_type_mismatch_raises_type_error(self) -> None:
236     with self.assertRaises(TypeError):
237         run_pipeline('x = 5; x = "hello";')
238
239 def test_undefined_variable_raises_type_error(self) -> None:
240     with self.assertRaises(TypeError):
241         run_pipeline("print(z);")
242
243 def test_function_wrong_arg_count_raises_type_error(self) -> None:
244     with self.assertRaises(TypeError):
245         run_pipeline("def f(a) { return a; } f(1, 2);")
246
247 def test_function_arg_type_mismatch_raises_type_error(self) -> None:
248     :
249     # first call infers a: Integer; second call with String must
250     fail
251     with self.assertRaises(TypeError):
252         run_pipeline('def f(a) { return a; } f(1); f("hello");')
253
254 def test_uncalled_function_body_type_error_is_caught(self) -> None:
255     # type error inside body must be detected even if function is
256     never called
257     with self.assertRaises(TypeError):
258         run_pipeline('def bad() { y = "hello" + 1; }')
259
260 def test_valid_uncalled_function_does_not_raise(self) -> None:
261     # a well-typed but uncalled function should not raise
262     result = run_pipeline("def add(a, b) { return a + b; }")
263     self.assertEqual(result.outputs, [])

```

```

259
260 #
261 -----
262 # Integration (full pipeline)
263 -----
264
265 class IntegrationTests(unittest.TestCase):
266     """End-to-end tests through all four pipeline stages."""
267
268     # --- Unary minus ---
269
270     def test_unary_minus_integer(self) -> None:
271         result = run_pipeline("x = -5; print(x);")
272         self.assertEqual(result.outputs, ["-5 : Integer"])
273
274     def test_unary_minus_float(self) -> None:
275         result = run_pipeline("y = -. 3.14; print(y);")
276         self.assertEqual(result.outputs, ["-3.14 : Float"])
277
278     def test_unary_minus_variable(self) -> None:
279         result = run_pipeline("x = 10; y = -x; print(y);")
280         self.assertEqual(result.outputs, ["-10 : Integer"])
281
282     def test_unary_minus_inside_expression(self) -> None:
283         result = run_pipeline("x = 3 + -2; print(x);")
284         self.assertEqual(result.outputs, ["1 : Integer"])
285
286     # --- Control flow ---
287
288     def test_if_then_branch_executes(self) -> None:
289         result = run_pipeline(
290             'x = 5; if (x == 5) { print("yes"); } else { print("no"); }
291             ,
292         )
293         self.assertEqual(result.outputs, ["yes" : String'])
294
295     def test_if_else_branch_executes_when_condition_false(self) -> None:
296         result = run_pipeline(
297             'x = 0; if (x == 5) { print("yes"); } else { print("no"); }
298             ,
299         )
300         self.assertEqual(result.outputs, ["no" : String'])
301
302     def test_while_loop_executes_repeatedly(self) -> None:
303         result = run_pipeline("x = 3; while (x != 0) { print(x); x = x
304             - 1; }")
305         self.assertEqual(result.outputs, ["3 : Integer", "2 : Integer",
306             "1 : Integer"])

```

```

303     # --- Functions ---
304
305     def test_function_integer_return(self) -> None:
306         result = run_pipeline("def square(n) { return n * n; } print(
307             square(7));")
308         self.assertEqual(result.outputs, ["49 : Integer"])
309
310     def test_function_type_inference(self) -> None:
311         result = run_pipeline(
312             "def add(a, b) { return a +. b; } result = add(2.0, 3.5);
313             print(result);"
314         )
315         self.assertEqual(result.outputs, ["5.5 : Float"])
316         self.assertIn("result          variable   Float", result.
317             checked_scope.format_table())
318
319     def test_function_value_parameter_passing(self) -> None:
320         # Modifying a parameter inside a function must not affect the
321         # caller's variable
322         result = run_pipeline(
323             "def inc(a) { a = a + 1; return a; } x = 10; y = inc(x);
324             print(x); print(y);"
325         )
326         self.assertEqual(result.outputs, ["10 : Integer", "11 : Integer
327             "])
328
329     def test_nested_function_calls(self) -> None:
330         result = run_pipeline(
331             "def add(a, b) { return a + b; } print(add(add(1, 2), 3));"
332         )
333         self.assertEqual(result.outputs, ["6 : Integer"])
334
335     # --- print() with all four types ---
336
337     def test_print_integer(self) -> None:
338         result = run_pipeline("print(42);")
339         self.assertEqual(result.outputs, ["42 : Integer"])
340
341     def test_print_float(self) -> None:
342         result = run_pipeline("print(3.14);")
343         self.assertEqual(result.outputs, ["3.14 : Float"])
344
345     def test_print_boolean(self) -> None:
346         result = run_pipeline("print(true);")
347         self.assertEqual(result.outputs, ["true : Boolean"])
348
349     def test_print_string(self) -> None:
350         result = run_pipeline('print("plc");')
351         self.assertEqual(result.outputs, ["'plc" : String'])
352
353     # --- Boolean expressions as values ---
354
355     def test_comparison_result_assigned_to_variable(self) -> None:

```

```

350     result = run_pipeline("x = 5; y = 5; flag = (x == y); print(
351         flag);")
352     self.assertEqual(result.outputs, ["true : Boolean"])
353
354     def test_comparison_result_printed_directly(self) -> None:
355         result = run_pipeline("x = 10; print(x != 0);")
356         self.assertEqual(result.outputs, ["true : Boolean"])
357
358     def test_comparison_result_false(self) -> None:
359         result = run_pipeline("a = 1; b = 2; eq = (a == b); print(eq);"
360             )
361         self.assertEqual(result.outputs, ["false : Boolean"])
362
363     # --- Reassignment ---
364
365     def test_variable_can_be_reassigned_same_type(self) -> None:
366         result = run_pipeline("x = 1; x = 2; x = 3; print(x);")
367         self.assertEqual(result.outputs, ["3 : Integer"])
368
369     def test_scope_isolation_if_block(self) -> None:
370         # Variable defined inside an if-block must not be visible
371         # outside it
372         with self.assertRaises(TypeError):
373             run_pipeline(
374                 "x = 1; if (x == 1) { inner = 99; } print(inner);"
375             )
376
377 if __name__ == "__main__":
378     unittest.main()

```

Listing 8.11: tests/test_pipeline.py — automated regression suite (62 tests)